Hidden Failures: Screening by Stress and the Value of Behavioral Data

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Abstract

In many risky jobs, workers can hide their failures. For example, a delivery rider who has a small crash can pay the other side privately and never report it. In this paper I study a platform that decides whether drivers work in a fast but risky mode (Express), and how intense this mode is. Drivers know privately if they can drive safely at high speed, the platform cannot see the care of the driver, a driver can hide an accident at cost κ, and payments cannot be negative (limited liability). I have four main results. First, when accidents are the only information, the Express market may not exist: care can be motivated only if $\kappa \geq c/(q_n - q_h)$. Even when the market exists, the platform must pay careless drivers a rent to keep them out of Express. Second, behavioral data from telematics, which cannot be hidden, makes the market possible. The optimal contract uses the most informative signals first. When hiding is cheap, it still pays drivers who report an accident but have a clean driving record. The platform does not choose this leniency, it is forced by hiding, and the platform would pay to make accidents harder to hide. Third, when the platform chooses the intensity, exposure to risk becomes a screening tool. A critical situation that is handled well looks the same as a calm trip, so under limited liability the rent goes down when exposure goes up (as long as passing a critical situation is the best-paid outcome). The platform therefore chooses more speed than is efficient, and drivers oppose speed targets. When the cost of care is proportional to exposure, this result reverses if the device also records exposure, and also under risk aversion without telematics. Fourth, in a two-period version, tests based on self-reported accidents lose their value when the stakes are high, but tests based on behavioral data do not. Speed caps and monitoring work as complements, because monitoring creates the markets where platforms over-speed.

Keywords: hidden failures, screening, moral hazard, costly falsification, limited liability, monitoring, telematics, platform work.
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1 Introduction

In many risky jobs, a failure does not always go on the record. A delivery rider who has a small crash can pay the other side and replace the order, and the platform never learns about it. A car driver can pay for a small accident by herself, so that she does not lose her no-claims discount. Boucher et al. (2009) estimate that about one quarter of the accidents in a Spanish insurance portfolio were never claimed. In aviation, non-punitive reporting systems were built because otherwise incidents are not reported (NASA Aviation Safety Reporting System, 2026). At the same time, there is now a new kind of data that records *how* people work, and not only what happened. Behavioral telematics records hard braking, speeding and phone use automatically. In a large U.S. monitoring program, drivers became about 30% safer while they were monitored, and safer drivers selected into monitoring (Jin and Vasserman, 2021).

These two features come together in platform work, where speed is now part of the product. Quick-commerce apps promise groceries in ten minutes, and delivery platforms rank riders by on-time performance. In January 2026, India's Ministry of Labour and Employment reportedly asked Blinkit, Swiggy Instamart and Zepto to stop advertising ten-minute delivery because of rider safety (TechCrunch, 2026). Later reports suggested that the pressure did not go away. It still comes through per-order pay, ratings, streak bonuses and automatic deactivation (Al Jazeera, 2026).

In this paper I ask what *hidden failures* do to the economics of risky work. I focus on four questions. Can a productive but risky activity exist at all if its failures can be hidden? What does the optimal contract look like? How much risk does a principal put on workers, when exposure to risk is also what shows who can handle it? And what should a regulator do about speed, monitoring data and accident reporting?

To answer these questions I use a simple model. A platform hires a driver, and the driver privately knows whether she is *careful-capable* (type H) or not (type L). The platform chooses if the driver works in a safe Standard mode or in a productive but risky Express mode. Later in the paper the platform also chooses how intense Express is. In Express, an H driver chooses whether to take costly care. An L driver cannot take care, because at high speed the cognitive load is too high for her (I call this a "cognitive wall"). The platform observes a telematics signal and the accident report of the driver. The driver can hide an accident at cost κ, and she can claim a fake accident for free. So a contract gives truthful reports only if the penalty for a reported accident is between 0 and κ. Payments satisfy limited liability. This is natural for gig workers, because a platform cannot charge them fees.

My first result is that hidden failures can make a risky market impossible. Without behavioral data, the only way to motivate care is a bonus for an accident-free record, and truthful reporting caps this bonus at κ. So care can be motivated only if $\kappa(q_n - q_h) \geq c$, where c is the cost of care and $q_n - q_h$ is how much care reduces the accident probability (Theorem 1). Below this threshold there is no Express market, no matter how productive it is. I call this an *implicit speed limit*. Above the threshold the market can still fail, because the platform must pay a rent to careless drivers to keep them out of Express. In Express they would get the accident-free bonus every time they are lucky.

I call this rent the *lemons tax*, and it is large exactly because accidents are rare.

My second result is that behavioral data creates the market, and that the leniency toward failure is forced and not chosen. With an informative telematics signal, the market can work for every $\kappa \geq 0$ (Theorem 2 and Corollary 2). I write each payment as a base bonus, which is paid after a telematics signal whether or not there is an accident, plus an accident-free premium, which is capped at κ. Then minimizing the rent becomes a fractional knapsack problem. The platform uses the instruments in decreasing order of their likelihood ratio, and the rent is the integral of the inverse informativeness along this pecking order. When hiding binds, the optimal contract pays drivers who report an accident but have a clean telematics record (Proposition 1). This leniency is not a tool of the platform. When accidents are bad news given each telematics signal, the optimal contract without hiding never pays after a reported accident, and the profit of the platform goes up strictly when failures become harder to hide, up to a threshold (Proposition 2). So the platform would like to punish failures more. It tolerates failures only because the driver can hide them.

My third result is that the platform screens by stress. When the platform chooses intensity, exposure to critical episodes is productive and also informative, because only in a critical episode do careful and careless drivers behave differently. A critical episode which is handled well leaves the same record as a calm trip. So at low intensity, a bonus for a clean record goes to careless drivers almost as often as to careful drivers. I show that in any limited-liability agency problem with this structure, where passing is the best-paid outcome, the rent goes down strictly with exposure (Theorem 3). Therefore the principal exposes the agent to more risk than is efficient. For the platform, this means that it runs faster than is efficient (Proposition 3). It also means that drivers have a direct stake in speed. The payoff of every driver goes down when the platform speeds up, so drivers oppose speed targets even when speed is productive (Proposition 4). This principle depends on what is recorded and on the friction. If the device also records exposure to critical episodes, the dilution goes away, and when the cost of care is proportional to exposure the distortion reverses (Proposition 6). With a risk-averse agent and no limited liability, the screening problem disappears. The moral-hazard contract already keeps the careless type out (Lemma 4), and with proportional care costs the platform chooses too little speed (Proposition 11).

My fourth result is about stress tests when failures can be hidden. In a two-period extension, Express works as a probation test for a valuable license. A test based on self-reported failures follows a version of Goodhart's law. When losing the license costs the driver more than κ, accident reports can be manipulated, and the value of the test goes to zero as the stakes grow (Proposition 13). A test based on behavioral data keeps its value at any stakes. So a platform with telematics may run a risky probation which loses money by itself, only to learn who can handle speed.

The model also gives some policy implications. First, speed caps and monitoring are *complements*. Monitoring creates the markets where the platform over-speeds. In these markets a moderate cap raises total surplus and moves surplus to drivers, but a cap that is too tight shuts down the market (Proposition 7). Second, a rule that forbids the use of telematics in pay can eliminate a risky market. This raises social welfare only when the harm of accidents to third parties

is larger than the private surplus (Proposition 8). Third, penalties for reported accidents that are above the hiding cost do not work, and investments that make failures harder to hide are substitutes for behavioral monitoring.

1.1 Relation to Chade and Swinkels (2026)

The closest paper is Chade and Swinkels (2026). They study a principal who wants a risk-averse agent to take a risky *initiative* instead of a safer project, and also to work hard. They give a likelihood-ratio condition under which pay is more convex than in the standard moral-hazard benchmark: low outputs are punished less, middle outputs are rewarded less, and high outputs are rewarded more. They also show by example that low or high outputs can be punished more instead, and that the principal tends to "go big or go home" on effort. Both papers are about risk and effort, but the two problems are different in each building block.

- *Who chooses the risk.* In their model the agent secretly chooses the risky project, and the principal must motivate her to take it. In my model the principal chooses the risky mode and its intensity, and contracts on them. The platform does not motivate the driver to take risk. It puts risk on her in order to learn about her.
- *Private information.* Their agent has no private information, and they list private information as an open question. Here the type of the driver is private. This creates the lemons tax and makes exposure to risk a screening device.
- *The failure signal.* In their model output is observed truthfully. Here failures can be hidden. This caps the penalty for accidents, can eliminate the market, and makes behavioral data valuable, because it cannot be manipulated. Their conclusion notes that the organizational responses they discuss involve changes in the information structure and need more modeling.
- *Leniency toward failure.* In their model, leniency toward failure, when it arises, is a tool that the principal chooses to make the initiative attractive, and they also show that it need not arise. Here leniency is forced by hiding, and the principal would pay to reduce it (Proposition 2). Also, since fake failures are free to claim, a contract here cannot pay more after a reported failure than after a success with the same behavioral record. In their model non-monotone contracts can be optimal.
- *Distortions.* They distort the effort of the agent, in either direction. Here the principal distorts the exposure to risk. It is distorted upward under limited liability with event-based monitoring, and the reason (clean records get diluted) has no counterpart in their model.

So I think the two papers are complementary. Their paper is about motivating an agent to take risk. This paper is about a principal who uses risk to learn about the agent, when the agent can hide the failures that the risk produces.

1.2 Related literature

This paper is related to four literatures.

First, the hiding constraint is a two-sided version of costly state falsification (Lacker and Weinberg, 1989). This literature shows that optimal contracts are falsification-proof, and that the cost of falsification bounds the slope of the contract. Related work studies insurance fraud (Crocker and Morgan, 1998) and earnings manipulation (Crocker and Slemrod, 2007). In law and economics, papers on self-reporting of harmful acts show that sanctions for self-reported violations must be moderate, otherwise people do not report (Kaplow and Shavell, 1994; Malik, 1993; Innes, 1999). I put the falsification bound into a problem with screening and moral hazard, where the principal chooses a risky technology. In this problem the bound creates a missing market, and together with a signal that cannot be manipulated it gives a pecking order and forced leniency.

Second, the paper is related to work on limited liability, informativeness and screening. Under limited liability the principal pays only after the most informative outcomes (Innes, 1990; Laffont and Martimort, 2002), and the value of information depends on likelihood ratios and not on the variance of the noise (Kim, 1995; Chaigneau et al., 2014). Ex post signals which are correlated with type reduce information rents (Riordan and Sappington, 1988; Demougin and Garvie, 1991; Gary-Bobo and Spiegel, 2006), and Martimort et al. (2025) study moral hazard, adverse selection and limited liability together. I add a falsification cap on one signal, and I let the allocation of the principal determine how informative the signals are. Georgiadis and Szentes (2020) study optimal monitoring design, Khalil and Lawarrée (1995) compare input and output monitoring, and Prat (2005) shows that in a career-concerns model, information about actions can be less valuable than information about consequences. In my model, information about actions (telematics) is what makes the market possible, because information about consequences (accidents) can be hidden. Barron et al. (2020) study agents who can add risk to output. Here it is the principal who chooses risk, and she does it to design information.

Third, the probation extension is related to test design under falsification (Perez-Richet and Skreta, 2022), to contracts for experimentation (Manso, 2011; Halac et al., 2016), and to career concerns and the informativeness of environments (Dewatripont et al., 1999). Manso (2011) shows that to motivate innovation the principal must tolerate early failure. There, like in Chade and Swinkels (2026), the tolerance is chosen when it arises. Here it is forced by hiding.

Fourth, the intensity margin is an activity level in the sense of Shavell (1980). Monitoring care does not discipline the activity level, and because of rent extraction the platform expands it too much. The result on aggregate accidents is similar to the offsetting behavior in Peltzman (1975), but here the mechanism is contractual and not behavioral. The literature on accident externalities (Edlin and Karaca-Mandic, 2006) motivates my welfare analysis. Papers on telematics in insurance study its welfare effects (Filipova-Neumann and Welzel, 2010) and document selection and behavioral responses (Jin and Vasserman, 2021; Reimers and Shiller, 2019).

The rest of the paper is organized as follows. Section 2 presents the model. Section 3 shows truthful reporting and reduces the problem to rent minimization. Section 4 derives the implicit

speed limit, the optimal contract with telematics, and forced leniency. Section 5 develops the stress-test principle and what it means for speed. Section 6 discusses policy. Section 7 studies risk aversion and Section 8 studies the probation extension. Section 9 concludes and sketches a continuous-time version. Proofs that are not in the text are in Appendix A.

2 Model

2.1 Environment

A risk-neutral principal (the platform, or an insurer) contracts with an agent (a driver). The driver has a type $\theta \in \{H, L\}$, which is her private information, and $\Pr(\theta = H) = \phi \in (0, 1)$. Type H is *careful-capable*: she can drive safely under pressure if she takes care. Type L cannot do this.

Modes. The driver works in one of two modes $K \in \{0, 1\}$, and the mode is contractible. In *Standard* mode ($K = 0$) she produces V_0 for the principal. Standard mode is safe: there are no accidents and nothing informative is recorded. In *Express* mode ($K = 1$) she produces $V_1 > V_0$, but accidents can happen. Each accident costs the principal $D \geq 0$ (for example liability, damaged goods, or brand and regulatory costs). The principal pays D whether the accident is reported or not. Reporting only changes what the contract can condition on.[1]

Care and the cognitive wall. In Express, an H driver chooses care $a \in \{0, 1\}$ at private cost $c\,a$, where $c > 0$. An L driver cannot take effective care, because driving safely at high speed is too hard for her. So there are three behavioral profiles in Express: h (H with care), n (H without care), and ℓ (type L).

2.2 Outcomes, information and reporting

An Express assignment gives an outcome $\omega = (y, s) \in \Omega := \{0, 1\} \times \mathcal{S}$. Here $y = 1$ means an accident, and s is a telematics signal with values in a finite set $\mathcal{S}$. Profile j gives a distribution $f_j \in \Delta(\Omega)$. Let

$$q_j := \textstyle\sum_s f_j(1, s), \qquad P_j(s) := f_j(0, s) + f_j(1, s)$$

be the accident probability and the marginal distribution of the signal under profile j.

Assumption 1 (Cognitive wall)**.** $f_\ell = f_n$: in Express, a type-L driver generates the same distribution of outcomes as a type-H driver who does not take care.

Assumption 1 is the idea that speed works as a stress test. In Express, the only thing that separates good records from bad records is whether the driver takes effective care. A careless type and a careful type who shirks cannot be distinguished statistically. Below I write n for both of them.

[1] If hiding also avoided the loss of the principal, it would be a (costly) private settlement and not a pure information problem. Section 6 discusses harm to third parties.

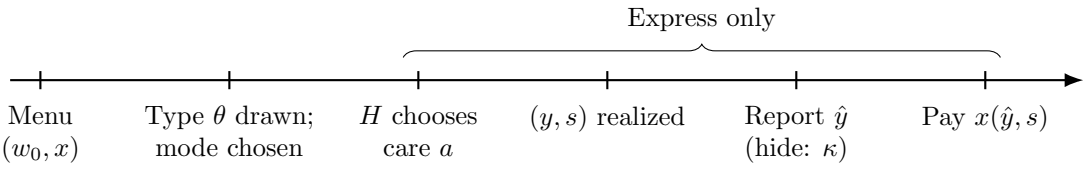


Figure 1: Timing.

Two monitoring regimes. *Without telematics*, $\mathcal{S}$ is a singleton and the only contractible Express outcome is the reported accident. *With telematics*, the signal is informative about care, $P_h \neq P_n$. In both regimes the accident technology is the same: $\sum_s f_j(y, s)$ does not depend on the regime.

Hiding and fabrication. The driver observes (y, s) and then sends a report $\hat{y} \in \{0, 1\}$. The telematics signal s is recorded automatically and cannot be manipulated. If $y = 1$, the driver can hide (conceal) the accident and report $\hat{y} = 0$ at cost $\kappa \geq 0$. For example, she can settle privately with a third party, repair the damage herself, or spend time and effort to keep the incident off the record. If $y = 0$, she can claim an accident, $\hat{y} = 1$, at no cost. Small incidents like a dropped order or a scratched vehicle are easy to claim.[2] Contracts can condition only on $(\hat{y}, s)$.

2.3 Contracts and timing

The agent is risk neutral, her outside option is normalized to 0, and she is protected by *limited liability*: all payments are nonnegative. A *menu* is a Standard wage $w_0 \geq 0$ together with an Express schedule $x : \{0, 1\} \times \mathcal{S} \to \mathbb{R}_+$, which gives the payment $x(\hat{y}, s)$.[3] The timing is as follows.

1. The principal offers a menu (w_0, x).
2. The driver learns θ and chooses Standard, Express, or her outside option.
3. In Express, an H driver chooses care a.
4. The outcome (y, s) is realized and the driver observes it.
5. The driver reports $\hat{y}$. She may hide an accident or fabricate one.
6. Payments are made, and the principal gets $V_K - Dy$.

As usual, a driver who is indifferent follows the recommendation of the principal.

Payoffs. If reporting is truthful, the expected payment to a driver with Express profile j is $W_j(x) := \sum_\omega f_j(\omega)\, x(\omega)$. Her payoff is $W_h(x) - c$ under profile h and $W_n(x)$ under profile n.

[2] Costless fabrication is a convenient benchmark, and it makes contracts weakly decreasing in reported accidents. In the binary-mode analysis of Section 4 it does not bind when accidents are bad news given each signal, $f_h(1, s)/f_n(1, s) \leq f_h(0, s)/f_n(0, s)$ for all s. It matters for the intensity analysis in Section 5.

[3] By the revelation principle, it is without loss to look at direct mechanisms that give each reported type a mode and a payment rule. Lemma 2 shows that the menu form is without loss. Lotteries over modes do not help, because all payoffs are linear in the assignment probabilities.

2.4 Assumptions and the first best

Assumption 2 (Care matters; careless Express is inefficient)**.** $q_h < q_n$, $S_n := V_1 - V_0 - q_n D < 0$, and $V_0 \geq 0$.

The first part says that care reduces accidents. The second part says that Express without care destroys value: the gain in productivity does not cover the expected accident cost of a careless driver. The third part says that Standard work is worth doing, so the principal never wants to leave a driver idle. I define the *Express surplus of a careful driver* as

$$S := V_1 - V_0 - q_h D - c.$$

If types and care were observable, the principal would assign H to Express with care whenever $S > 0$, assign L to Standard, and pay each driver her cost. The first-best total surplus is $V_0 + \phi \max\{S, 0\}$. Under Assumption 2, Express can be efficient only for careful drivers, and only if they really take care.

2.5 A running example

Some results have a simple form when telematics is binary. Let $\mathcal{S} = \{G, B\}$ ("good" and "bad" driving behavior), let $\pi_j := P_j(B)$ with $\pi_h < \pi_n$, and assume that accidents and signals are independent conditional on the profile, $f_j(y, s) = \Pr_j(y) P_j(s)$, where $\Pr_j(1) = q_j$ and $\Pr_j(0) = 1 - q_j$. The example includes the no-telematics regime as the case $\pi_h = \pi_n$. Figure 2 below uses $q_h = 0.05$, $q_n = 0.15$, $\pi_h = 0.1$, $\pi_n = 0.3$, $c = 1$, and $S = 10$.

3 Truthful reporting and the lemons tax

3.1 Truthful reporting is without loss

Let

$$\mathcal{X}(\kappa) := \left\{ x \in \mathbb{R}_+^{\Omega} \,:\, 0 \leq x(0, s) - x(1, s) \leq \kappa \;\text{ for all } s \in \mathcal{S} \right\}.$$

A schedule in $\mathcal{X}(\kappa)$ never rewards a reported accident, so the driver has no incentive to fabricate. It also never penalizes a reported accident by more than κ, so the driver has no incentive to hide.

Lemma 1 (Truthful reporting)**.** *For any Express schedule $x \geq 0$, define*

$$\hat{x}(0, s) := \max\{x(0, s),\, x(1, s)\}, \qquad \hat{x}(1, s) := \max\{x(1, s),\, x(0, s) - \kappa\}.$$

Then (i) $\hat{x} \in \mathcal{X}(\kappa)$; (ii) at every outcome (y, s), the payoff of the driver under $\hat{x}$ with truthful reporting is equal to her payoff under x with optimal reporting; and (iii) at every outcome, the payment of the principal under $\hat{x}$ is weakly lower, and strictly lower when an accident would be hidden under x and $\kappa > 0$. So it is without loss to restrict attention to schedules in $\mathcal{X}(\kappa)$, and in equilibrium every accident is reported.

The idea is the same as in Lacker and Weinberg (1989). Hiding only wastes resources. So the principal can directly promise the driver what she would get by hiding, and save the hiding cost. The payoff of the driver at each outcome does not change, so the incentives for care and for self-selection also do not change. From now on, contracts are elements of $\mathcal{X}(\kappa)$.

3.2 Reduction to a rent-minimization problem

For a given κ, I define the *rent function*

$$R(\kappa) := \min\Big\{W_n(x) \,:\, x \in \mathcal{X}(\kappa),\ W_h(x) - W_n(x) \geq c\Big\}, \tag{1}$$

with $R(\kappa) := +\infty$ if the constraint set is empty. The constraint $W_h - W_n \geq c$ is the care (moral-hazard) constraint. The objective $W_n(x)$ is what a careless driver, of either type, expects to get from the Express schedule.

Lemma 2 (Reduction)**.** *Under Assumptions 1 and 2, the maximal expected profit of the principal is*

$$\Pi^* = V_0 + \max\big\{0,\ \phi S - R(\kappa)\big\}.$$

If $\phi S > R(\kappa)$, an optimal menu is (w_0^*, x^*), where x^* solves (1) with a binding care constraint and $w_0^* = W_n(x^*)$. Type H chooses Express and takes care, type L chooses Standard, and each *type gets rent $R(\kappa)$. If $\phi S < R(\kappa)$, it is optimal to assign both types to Standard at wage zero.*

Proof sketch. The full proof is in Appendix A. Suppose the principal induces care from H in Express with schedule x. Then L can also take this schedule and behave as profile n. So L must get at least $W_n(x)$ wherever she is assigned. Since it is inefficient to put L in Express (Assumption 2), the best place for her is Standard with $w_0 = W_n(x)$. The care constraint gives $W_h(x) \geq c + W_n(x)$, so H also gets at least $W_n(x)$. Adding up, the profit of the principal is at most $V_0 + \phi S - W_n(x) \leq V_0 + \phi S - R(\kappa)$, and the menu above achieves this bound. Any mechanism that does not induce care gives at most V_0. □

Remark 1 (The lemons tax). The rent $R(\kappa)$ is paid to *every* driver. For type H it is the usual limited-liability rent from moral hazard, and she would get it even without adverse selection. The cost of adverse selection is the extra rent $(1-\phi)R(\kappa)$ paid to careless types, who never produce Express surplus. It is paid through the Standard wage: slow work must pay enough so that careless drivers do not want to gamble on the Express schedule. It is optimal to operate Express only if

$$\phi S \ \geq\ R(\kappa),$$

while if careless drivers could be identified and excluded the condition would be $S \geq R(\kappa)$. When there are more careless drivers, the high-speed market is harder to sustain, like in Akerlof (1970). The extra rent is the price of keeping the "lemons" in the slow lane.

Remark 2 (Why limited liability matters). Under the cognitive wall, the payoff of L from mimicking H is the same as the payoff of H from shirking. With limited liability, this payoff is strictly positive whenever care is motivated, unless some outcome is possible only for careful drivers (in that case R can be zero). So screening is costly. Section 7 shows what happens without limited liability, when the agent is risk averse and can be held to her reservation utility. Then the mimicking payoff of the careless type goes down to her outside option, and screening costs nothing more than the moral-hazard problem.

4 The implicit speed limit and the value of behavioral data

4.1 Accidents only

Theorem 1 (Implicit speed limit). *Suppose there is no telematics ($\mathcal{S}$ is a singleton), and let*

$$\kappa_0 := \frac{c}{q_n - q_h}.$$

(i) Care in Express is implementable if and only if $\kappa \geq \kappa_0$.

(ii) If $\kappa \geq \kappa_0$, the rent is $R_0 := c(1-q_n)/(q_n-q_h)$. It is attained by paying κ_0 for an accident-free record and 0 after a reported accident. (For $\kappa < \kappa_0$ the rent is $+\infty$. I write $R_0(\kappa)$ for the rent as a function of κ.)

(iii) Operating Express is optimal if and only if $\kappa \geq \kappa_0$ and $\phi S \geq R_0$. In particular, for $\kappa < \kappa_0$ Express is shut down, no matter how large S is.

Proof. With a singleton signal, let $d := x(0) - x(1) \in [0, \kappa]$. Then $W_j(x) = x(1) + (1 - q_j)\, d$, so the care constraint is $(q_n - q_h)d \geq c$, that is, $d \geq \kappa_0$. This is feasible iff $\kappa \geq \kappa_0$. The rent $W_n = x(1) + (1 - q_n)d$ is minimized at $x(1) = 0$ and $d = \kappa_0$. Part (iii) follows from Lemma 2. □

The threshold κ_0 is the smallest accident-free bonus that makes care worthwhile. If a driver can hide an accident for less than this, then no scheme based on accidents can motivate care. Any penalty that is large enough to matter is also large enough to be avoided by hiding. So the problem comes from the *interaction* of moral hazard and falsification, and adverse selection enters through the rent. I think the speed limit is relevant in practice for two reasons. First, accidents are rare, so $q_n - q_h$ is small and κ_0 is large. For example, if care lowers the accident probability per shift from 3% to 1%, the hiding cost must be at least fifty times the cost of care. Second, the incidents that are cheapest to hide (small collisions, damaged goods, near-injuries) are exactly the ones that happen often enough to be informative.

The rent $R_0 \approx c/(q_n - q_h)$ is large for the same reason. A careless driver gets the accident-free bonus with probability $1 - q_n$, which is close to one. Figure 2 shows both problems. For $\kappa < \kappa_0$ no contract exists. For $\kappa \geq \kappa_0$ the market survives only if $\phi S \geq R_0$, and in the figure this fails when $\phi = 0.6$.

4.2 The optimal contract with telematics

Any $x \in \mathcal{X}(\kappa)$ can be written in a unique way as

$$x(1,s) = b_s \geq 0, \qquad x(0,s) = b_s + d_s, \quad d_s \in [0,\kappa],$$

where b_s is a *base bonus*, paid after signal s whether or not an accident is reported, and d_s is an *accident-free premium*. I think of each of them as an "instrument" with an incentive yield, a rent cost and a likelihood ratio:

instrument	yield m	rent cost ρ	likelihood ratio	capacity
base bonus b_s	$P_h(s) - P_n(s)$	$P_n(s)$	$\lambda_s := P_h(s)/P_n(s)$	∞
accident-free premium d_s	$f_h(0,s) - f_n(0,s)$	$f_n(0,s)$	$\mu_s := f_h(0,s)/f_n(0,s)$	κ

Indeed, $W_h(x) - W_n(x) = \sum_s [m(b_s)b_s + m(d_s)d_s]$ and $W_n(x) = \sum_s [\rho(b_s)b_s + \rho(d_s)d_s]$. For every instrument, $m/\rho = \mathrm{LR} - 1$. Let $\lambda^* := \max_s \lambda_s$. Since P_h and P_n are both probability distributions, $\lambda^* \geq 1$, with equality iff the signal is uninformative about care.

Theorem 2 (Pecking order)**.** *The rent problem* (1) *is the fractional knapsack*

$$R(\kappa) = \min_z \sum_i \rho_i z_i \quad s.t. \quad \sum_i m_i z_i \geq c, \quad 0 \leq z_i \leq \mathrm{cap}_i,$$

over the $2|\mathcal{S}|$ *instruments above. It is solved by* filling the instruments with $m_i > 0$ in decreasing order of their likelihood ratio *until the care constraint binds. As a result:*

(i) Rent formula. *Whenever* $R(\kappa) < \infty$, $R(\kappa) = \int_0^c \frac{dt}{\Lambda(t)-1}$, *where* $\Lambda(t)$ *is the likelihood ratio of the instrument that is marginal when the cumulative incentive yield is equal to* t *(with* $1/(\infty - 1) := 0$*). R is nonincreasing and convex in* κ*.*

(ii) Telematics. *If* $\lambda^* > 1$*, then* $R(\kappa) < \infty$ *for every* $\kappa \geq 0$*. There is an optimal schedule that pays accident-free premia only on signals in* $\mathcal{S}^+ := \{s : \mu_s > \lambda^*\}$*, filled in decreasing order of* μ_s *up to* κ *each. It pays a base bonus only on a best signal* $s^* \in \arg\max_s \lambda_s$*, and only if the premia at capacity do not satisfy the care constraint.*

(iii) No telematics. *If* $\lambda^* = 1$*, then* $R(\kappa) < \infty$ *iff* $\kappa \sum_{s:\mu_s>1} [f_h(0,s) - f_n(0,s)] \geq c$*. With a singleton signal this becomes* $\kappa \geq \kappa_0$*.*

The logic of the pecking order is simple. Each dollar of incentive $W_h - W_n$ that is bought with instrument i leaks $1/(\mathrm{LR}_i - 1)$ dollars of rent W_n to careless drivers. So the platform uses the most discriminating outcomes first. Usually the most discriminating outcome is "no accident *and* a clean telematics record", but its use is capped by κ. When the cap binds, the platform moves down to the best instrument that cannot be manipulated, the telematics base bonus, which has unlimited capacity. Without an informative signal there is no instrument with unlimited capacity, and the cap gives the speed limit.

When $\mathcal{S}^+$ is nonempty, let

$$\kappa_T := \frac{c}{\sum_{s\in\mathcal{S}^+}[f_h(0,s) - f_n(0,s)]}$$

be the hiding cost above which the accident-free premia alone can give the care incentive. For $\kappa < \kappa_T$ the hiding constraint binds and a base bonus is needed. For $\kappa \geq \kappa_T$ the premia alone are enough. But the premia on the most informative signals may stay at the cap, and the rent may continue to fall, until κ reaches $c/[f_h(0,s_1) - f_n(0,s_1)]$ for the signal $s_1 \in \mathcal{S}^+$ with the highest μ_s. In the binary example with $\mu_B \leq \lambda_G$ (Corollary 1) the two thresholds are the same.

Corollary 1 (Binary telematics)**.** *In the running example, suppose $\mu_B \leq \lambda_G$, that is, $\pi_h(1-q_h)/[\pi_n(1-q_n)] \leq (1-\pi_h)/(1-\pi_n)$. Let $m_G := (1-\pi_h)(1-q_h) - (1-\pi_n)(1-q_n)$. Then $\mathcal{S}^+ = \{G\}$, so $\kappa_T = c/m_G$; an optimal schedule sets $x(0,B) = x(1,B) = 0$; and*

$$R_T(\kappa) = \begin{cases} \dfrac{1-\pi_n}{\pi_n - \pi_h}\left[c - (1-\pi_h)(q_n - q_h)\,\kappa\right], & \kappa < \kappa_T, \\ \dfrac{c\,(1-\pi_n)(1-q_n)}{m_G}, & \kappa \geq \kappa_T. \end{cases}$$

For $\kappa \geq \kappa_T$ the schedule pays κ_T after $(0,G)$ and nothing otherwise. For $\kappa < \kappa_T$ it pays $x(1,G) = m_G(\kappa_T - \kappa)/(\pi_n - \pi_h) > 0$ and $x(0,G) = x(1,G) + \kappa$.

Proof. Here $\lambda^* = \lambda_G = (1-\pi_h)/(1-\pi_n) > 1$ and $\mu_G = \lambda_G(1-q_h)/(1-q_n) > \lambda^*$, while $\lambda_B < 1$ and $\mu_B \leq \lambda^*$. So the order is d_G (capacity κ), then b_G. If $\kappa\, m_G \geq c$, set $d_G = c/m_G = \kappa_T$. Otherwise set $d_G = \kappa$ and $b_G = (c - \kappa m_G)/(\pi_n - \pi_h)$. The rent $\kappa(1-\pi_n)(1-q_n) + b_G(1-\pi_n)$ simplifies to the expression above. □

Proposition 1 (Accident forgiveness)**.** *Suppose $\lambda^* > 1$, $\mathcal{S}^+$ is nonempty, every outcome has positive probability under both profiles, all likelihood ratios are distinct, and $\kappa < \kappa_T$, so that the hiding constraint binds. Then the optimal schedule is unique and*

(i) penalizes a reported accident by exactly κ after every signal in $\mathcal{S}^+$, and by 0 after every other signal;

(ii) pays a strictly positive amount after a reported accident together with the best signal: $x(1,s^*) = b_{s^*} > 0$;

(iii) as $\kappa \downarrow 0$, converges to a pure telematics bonus with no accident penalty, $x(0,s^*) = x(1,s^*) = c/[P_h(s^*) - P_n(s^*)]$.

Proof. Since $\kappa < \kappa_T$, the premia on $\mathcal{S}^+$ at capacity do not satisfy the care constraint. So by Theorem 2(ii) all of them are at κ, and the rest is covered by $b_{s^*} > 0$. Uniqueness follows because the greedy order is strict. Part (iii) follows by letting $\kappa \to 0$ in $b_{s^*} = (c - \kappa\sum_{\mathcal{S}^+} m(d_s))/m(b_{s^*})$. □

4.3 Leniency is forced, not chosen

The forgiveness here is not a choice of the platform to be lenient. The platform would like to punish accidents more, because they are informative. But a harder punishment would only push accidents underground. The next result shows this formally.

Proposition 2 (Leniency is forced, not chosen)**.** *Suppose telematics is informative ($\lambda^* > 1$), every outcome has positive probability under both profiles, all likelihood ratios are distinct, and accidents are bad news given every signal: $\mu_s > \lambda_s$ for all s. Then $\mathcal{S}^+$ is nonempty, so $\kappa_T < \infty$, and:*

(i) If $\kappa \geq \kappa_T$, and in particular if accidents cannot be hidden at all, the optimal contract pays nothing after any reported accident.

(ii) If $\kappa < \kappa_T$, the optimal contract pays a positive amount after a reported accident together with the best signal (Proposition 1).

(iii) Let $s_1 \in \arg\max_s \mu_s$ and $\kappa^{\text{top}} := c/[f_h(0, s_1) - f_n(0, s_1)] \geq \kappa_T$. The rent $R(\kappa)$ is strictly decreasing on $[0, \kappa^{\text{top}})$ and constant on $[\kappa^{\text{top}}, \infty)$. So whenever Express operates, the profit of the platform goes up strictly when failures become harder to hide, up to κ^{top}.

In the running example with $\mu_B \leq \lambda_G$ (Corollary 1), $\kappa^{\text{top}} = \kappa_T$, and each unit by which κ is below κ_T costs the platform $(1 - \pi_n)(1 - \pi_h)(q_n - q_h)/(\pi_n - \pi_h)$ in rent.

Proof. Since $\mu_{s^*} > \lambda_{s^*} = \lambda^*$, the best signal is in $\mathcal{S}^+$, so $\mathcal{S}^+$ is nonempty, and $s_1 \in \mathcal{S}^+$ because $\mu_{s_1} \geq \mu_{s^*}$. (i) If $\kappa \geq \kappa_T$, the premia on $\mathcal{S}^+$, whose likelihood ratios are higher than that of every base bonus, give the care incentive before any base bonus is reached in the pecking order of Theorem 2. So $b_s = x(1, s) = 0$ for all s, and the optimum is unique because the order is strict. (ii) is Proposition 1. (iii) For $\kappa \geq \kappa^{\text{top}}$ the premium on s_1 alone gives the incentive, so the optimal contract and the rent $c\, f_n(0, s_1)/[f_h(0, s_1) - f_n(0, s_1)]$ do not depend on κ. For $\kappa < \kappa^{\text{top}}$ the premium on s_1 is at its cap, and the marginal instrument has a strictly lower likelihood ratio. If the cap goes up by ε, a yield of $\varepsilon[f_h(0, s_1) - f_n(0, s_1)]$ can be moved from the marginal instrument to d_{s_1}, and by the rent formula of Theorem 2(i) this strictly lowers the rent. The slope in the running example follows from Corollary 1. □

This is the opposite of the logic of tolerance for failure in Manso (2011) and Chade and Swinkels (2026). In these papers, leniency toward failure, when it is optimal, is chosen by the principal to make a risky choice attractive to the agent, so it is an instrument. Here the platform chooses the risky mode itself, and leniency is a constraint that it would pay to relax. The platform could relax it by investing in dashcams, police reports or other ways to verify outcomes, which raise κ. So outcome verification and behavioral monitoring are substitutes: the first makes failures harder to hide, and the second gives evidence that cannot be hidden. "Accident forgiveness" clauses in auto insurance and the non-punitive incident reporting in aviation safety[4] have the same logic as

[4]Under NASA's Aviation Safety Reporting System, if a person files a report within ten days of an inadvertent violation, she can qualify for a waiver of FAA penalties. Accidents and criminal acts are excluded (NASA Aviation Safety Reporting System, 2026).

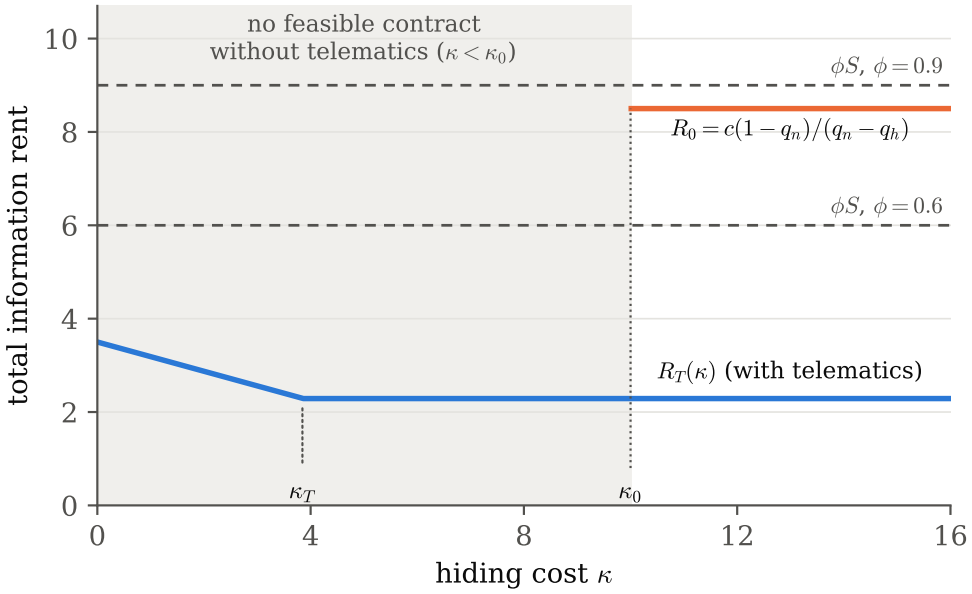


Figure 2: Rents with and without telematics as functions of the hiding cost ($q_h = 0.05$, $q_n = 0.15$, $\pi_h = 0.1$, $\pi_n = 0.3$, $c = 1$, $S = 10$). Without telematics, no contract motivates care for $\kappa < \kappa_0 = 10$, and for $\kappa \geq \kappa_0$ the rent is $R_0 = 8.5$. With telematics, the rent falls linearly in κ until the accident-free premium stops binding at $\kappa_T \approx 3.85$. The market operates where the rent is below ϕS: without telematics only for $\kappa \geq 10$ and $\phi = 0.9$, and with telematics for all κ and both values of ϕ.

Proposition 1: when failures can be hidden, the penalty for reporting one cannot be higher than the cost of hiding it.

4.4 Market creation

Let $R_0(\kappa)$ and $R_T(\kappa)$ be the rent without and with telematics.

Corollary 2 (Telematics creates the market)**.** *$R_T(\kappa) \leq R_0(\kappa)$ for all κ. If telematics is informative, $R_T(\kappa) < \infty$ for all $\kappa \geq 0$. So operating Express is optimal with telematics whenever it is optimal without telematics, and also*

- *on $\{\kappa < \kappa_0,\ \phi S \geq R_T(\kappa)\}$, where telematics* creates *the market; and*
- *on $\{\kappa \geq \kappa_0,\ R_T(\kappa) \leq \phi S < R_0\}$, where telematics* relieves the lemons tax.

On both sets, total surplus goes up by ϕS.

Proof. A schedule that depends only on $\hat{y}$ is feasible with telematics and gives the same W_h and W_n, because the accident marginals do not depend on the regime. So $R_T \leq R_0$. Finiteness follows from Theorem 2(ii), and the rest follows from Lemma 2. □

Figure 2 shows this. Telematics turns a high-speed market that is infeasible or unprofitable into a profitable one. It does this not by improving an existing contract, but by adding an instrument with unlimited capacity. This is different from the static insurance literature. In Rothschild and

Stiglitz (1976) and Stiglitz (1977), better information refines prices in markets that already exist. Here, the market does not exist without the right kind of information.

5 Screening by stress

Until now the risky mode was a single option. But platforms also choose *how hard* Express runs, for example through delivery-time targets, orders per hour or routing density. Exposure to risk is productive and costly, but it is also informative, because careful and careless drivers behave differently only in critical situations. In this section I show that under limited liability, a principal who can choose exposure uses it as a screening device, and she exposes the agent to more risk than is efficient. I first state the principle for a general agency problem, and then I apply it to speed.

5.1 A general stress-test principle

Consider any agency problem with the following structure. A *critical episode* produces an outcome in a finite set Ω, with distribution f_h if the agent is careful and f_n if she is not. One outcome $\omega_0 \in \Omega$, the *pass* outcome, is also what the principal records when no critical episode happens. The principal chooses an exposure $k \in (0,1]$, which is the probability that the agent faces a critical episode, and she does not observe whether one happened. So outcomes are distributed according to

$$f_j^k := k\, f_j + (1-k)\, \mathbf{1}_{\omega_0}, \qquad j \in \{h, n\}.$$

By Blackwell (1953), the experiment at exposure $k' > k$ is more informative than the experiment at k: we get f_j^k from $f_j^{k'}$ by replacing each outcome with ω_0 with probability $1 - k/k'$. The admissible contracts form a polyhedron $\mathcal{X} \subseteq \mathbb{R}_+^\Omega$ that does not depend on k (for example, limited liability together with reporting constraints). Being careful costs $c(k) > 0$, and $c(k)/k$ is nonincreasing. Let $W_j^k(x) := \sum_\omega f_j^k(\omega) x(\omega)$, and let

$$R(k) := \min\big\{W_n^k(x) : x \in \mathcal{X},\ W_h^k(x) - W_n^k(x) \geq c(k)\big\}, \qquad \mathcal{K} := \{k \in (0,1] : R(k) < \infty\}.$$

Suppose also that the setting has the two-type structure of Section 2 (the outcomes of the careless type are distributed as f_n, careless work is inefficient at every exposure, and the safe alternative is worth $V_0 \geq 0$) and that $0 \in \mathcal{X}$. Then Lemma 2 applies at every k: every agent gets $R(k)$, and the profit of the principal is $V_0 + \max\{0, \phi S(k) - R(k)\}$, where $S(k)$ is the surplus of careful work at exposure k.

Definition 1. *Pass is the top outcome* if, at every $k \in \mathcal{K}$, some solution x^k of the rent problem pays weakly more after ω_0 than after any other outcome: $x^k(\omega_0) = \max_\omega x^k(\omega)$.

Theorem 3 (Stress-test principle)**.** *Suppose pass is the top outcome.*

(i) $\mathcal{K}$ is an up-set, and R is strictly decreasing on $\mathcal{K}$.

(ii) If the maximizers below exist, every second-best exposure $k^{\mathrm{SB}} \in \arg\max_{k\in\mathcal{K}}[\phi S(k) - R(k)]$ *is weakly larger than every first-best exposure* $k^{\mathrm{FB}} \in \arg\max_k S(k)$.

(iii) If in addition k^{FB} *is interior and* S *is differentiable at* k^{FB}*, then* $k^{\mathrm{SB}} > k^{\mathrm{FB}}$.

Proof. (i) Since the pass component is the same for both behaviors, $W_h^k(x) - W_n^k(x) = k[W_h(x) - W_n(x)]$ and $W_n^k(x) = kW_n(x) + (1-k)x(\omega_0)$. A contract that satisfies the care constraint at k satisfies $W_h(x) - W_n(x) \geq c(k)/k \geq c(k')/k'$ for every $k' > k$, so it is feasible at k'. Therefore $\mathcal{K}$ is an up-set. Fix $k < k'$ in $\mathcal{K}$ and let $x = x^k$. Its rent at k' is

$$W_n^{k'}(x) = R(k) - (k'-k)\,\Theta, \qquad \Theta := x(\omega_0) - W_n(x) = \sum_\omega f_n(\omega)\big[x(\omega_0) - x(\omega)\big] \;\geq\; 0.$$

If $\Theta = 0$, then $x(\omega) = \max x$ for every ω in the support of f_n, so $W_n(x) = \max x \geq W_h(x)$. This contradicts $W_h(x) - W_n(x) \geq c(k)/k > 0$. So $\Theta > 0$ and $R(k') \leq R(k) - (k'-k)\Theta < R(k)$.

(ii) Suppose $k^{\mathrm{SB}} < k^{\mathrm{FB}}$. Since $\mathcal{K}$ is an up-set, $k^{\mathrm{FB}} \in \mathcal{K}$. Optimality gives $\phi S(k^{\mathrm{SB}}) - R(k^{\mathrm{SB}}) \geq \phi S(k^{\mathrm{FB}}) - R(k^{\mathrm{FB}})$ and $S(k^{\mathrm{FB}}) \geq S(k^{\mathrm{SB}})$. Adding them, we get $R(k^{\mathrm{SB}}) \leq R(k^{\mathrm{FB}})$, which contradicts (i).

(iii) If $k^{\mathrm{FB}} \notin \mathcal{K}$, then every element of the up-set $\mathcal{K}$, and in particular k^{SB}, is larger than k^{FB}. Otherwise, the bound in (i) with $k = k^{\mathrm{FB}}$ gives, for k' slightly above k^{FB},

$$\phi S(k') - R(k') \;\geq\; \phi S(k^{\mathrm{FB}}) - R(k^{\mathrm{FB}}) + (k' - k^{\mathrm{FB}})\,\Theta + o(k' - k^{\mathrm{FB}}),$$

because $S'(k^{\mathrm{FB}}) = 0$. The right-hand side is larger than $\phi S(k^{\mathrm{FB}}) - R(k^{\mathrm{FB}})$ for k' close to k^{FB}, so k^{FB} is not optimal, and by (ii), $k^{\mathrm{SB}} > k^{\mathrm{FB}}$. □

There are two forces behind the theorem. The first one is a *dilution effect*. The principal must reward the pass outcome, because passing is what care produces in a critical episode. But passing is also what every agent produces when there is no critical episode. At low exposure most passes are uninformative, so the reward goes to careless agents almost as often as to careful ones, and it leaks as rent. When exposure goes up, passes become rarer for the careless agent and so more informative, and the rent goes down. The second force is a *leverage effect*, which is present when the cost of care does not scale with exposure. If care costs c but matters only in a fraction k of trips, then the per-episode incentive $c(k)/k$ that the contract must give is larger at low exposure. Under limited liability this raises the rent only by making the dilution stronger or when a cap on payments binds. Under risk aversion it always raises the agency cost, through the risk premium (Section 7). With a proportional cost, $c(k) = k\gamma$, there is no leverage effect and only dilution drives the theorem.

The principle is not only about driving. It applies whenever a principal decides how much of the work of an agent happens in situations that reveal competence, and when handling such a situation well leaves the same record as not facing it at all. Some examples are a new employee who gets demanding accounts, a trainee who gets difficult cases, or a courier who gets tight delivery

windows. Screening pushes the principal toward *more* exposure than is efficient. This is the opposite of the downward distortion that we know from screening models (Mussa and Rosen, 1978). Three conditions are important. The first one is limited liability: under risk aversion, with proportional care costs, the direction can reverse (Proposition 11). The second one is that the record does not reveal exposure. If it does, the pass outcome is not pooled with uneventful work anymore, and the dilution effect disappears (Proposition 6). The third one is that passing is the best-paid outcome. If the contract can reward an outcome that happens only under exposure (a good-news event, or an accident when fabrication is costly), these payments are not diluted (Remark 3).

5.2 Application: speed

In the driving model, a critical episode is a tight turn, a sudden obstacle, or a rushed intersection. Intensity k is the probability that an Express trip has one, and it is contractible, for example as a delivery-time target. Outside critical episodes there is no accident and no telematics event. Let $s_0 \in \mathcal{S}$ be the "no event" signal ($s_0 = G$ in the running example), so the pass outcome is $\omega_0 := (0, s_0)$, and let $\mathcal{X} = \mathcal{X}(\kappa)$. I call this the *critical-episode model*. Output is $V(k)$, the accident probability of profile j is kq_j, and the surplus of a careful driver is $S(k) := V(k) - V_0 - kq_hD - c(k)$. I keep κ fixed, and with a small abuse of notation I write $R(k)$ for the rent at intensity k and $\mathcal{K}(\kappa)$ for the set of implementable intensities.

Assumption 3. (i) $c(k) > 0$ and $c(k)/k$ is nonincreasing, so the cost of care grows at most proportionally with exposure; (ii) careless Express is inefficient at every intensity, $V(k) - V_0 - kq_nD < 0$; and (iii) telematics events are bad news: for every $s \neq s_0$, $\lambda_s \leq 1$ and $\mu_s \leq 1$.

Part (i) includes a fixed cost of attention ($c(k) = c$) and also a cost proportional to critical episodes ($c(k) = k\gamma$). Part (ii) extends Assumption 2, and it makes Lemma 2 apply at every intensity. So the profit of the principal at intensity k is $V_0 + \max\{0, \phi S(k) - R(k)\}$. Part (iii) says that an event signal $s \neq s_0$, alone or together with no accident, does not raise the likelihood that the driver was careful. In the running example it holds iff $\pi_h(1 - q_h) \leq \pi_n(1 - q_n)$, and without telematics it holds trivially.

Lemma 3 (Implementability at intensity k)**.** *Care at intensity k is implementable iff $c(k)/k \leq \bar{Y}(\kappa) := \sup\{W_h(x) - W_n(x) : x \in \mathcal{X}(\kappa)\}$, where $\bar{Y}(\kappa) = \infty$ with informative telematics and $\bar{Y}(\kappa) = \kappa(q_n - q_h)$ without telematics. Under Assumption 3(i), $\mathcal{K}(\kappa)$ is an up-set.*

Proof. By the identities in the proof of Theorem 3(i), implementability is the same as $W_h(x) - W_n(x) \geq c(k)/k$ for some $x \in \mathcal{X}(\kappa)$. The values of $\bar{Y}$ follow from Theorem 2. With telematics, the base bonus on s^* has unlimited capacity. Without telematics, $W_h - W_n = (q_n - q_h)d \leq (q_n - q_h)\kappa$. The monotonicity of $c(k)/k$ gives the up-set property. □

Proposition 3 (The platform over-speeds)**.** *Under Assumptions 1 to 3, with or without telematics, pass is the top outcome in the critical-episode model. At every k there is an optimal schedule that pays only after the no-event signal, $x(0, s_0) = b + d$ and $x(1, s_0) = b$ with $d \in [0, \kappa]$ and $b \geq 0$.*

So the conclusions of Theorem 3 hold: the rent strictly falls with intensity, and whenever Express operates, the platform runs it at an intensity weakly above the efficient one (and strictly above under the conditions of part (iii)).

Proof. Contracts in $\mathcal{X}(\kappa)$ never reward a reported accident (because fabrication is costless), so after each signal s the only instruments are the base bonus b_s and the premium d_s of Section 4. By Assumption 3(iii), every instrument on a signal $s \neq s_0$ has a nonpositive incentive yield and a nonnegative rent cost at every intensity, so it can be set to zero without loss. The remaining schedule pays $b + d$ after ω_0, b after $(1, s_0)$, and nothing otherwise, so pass is the top outcome. □

Figure 3(a) plots the first-order conditions. The second-best intensity sets the marginal surplus $\phi S'(k)$ equal to $R'(k) < 0$, so it is to the right of k^{FB}.

Remark 3 (The role of the assumptions). Costless fabrication is important. Without it, a contract could pay after an accident together with a clean signal. This outcome happens only in critical episodes, so it is not diluted, and then the rent can go up with intensity. Both parts of Assumption 3(iii) are important for the same reason: if an event signal, or an event signal with no accident, were good news, the platform would pay on it, and such payments are also not diluted. In each of these cases pass is no longer the top outcome. Finally, the rent does not go to zero when telematics becomes perfect. Even if a careless driver always produces an event in a critical episode, calm trips still give clean records for everyone. So the rent is $(1 - k)$ times the bonus, and the platform still wants to raise intensity.

5.3 Drivers' stake in speed

The stress-test principle also predicts who wants speed.

Proposition 4 (Drivers oppose speed targets)**.** *Under the assumptions of Proposition 3, when Express operates at intensity k, the equilibrium payoff of each driver, of either type, is equal to $R(k)$, which is strictly decreasing in k. So:*

(i) every driver strictly prefers any lower intensity at which Express still operates;

(ii) whenever Express is viable at the efficient intensity, $\phi S(k^{\mathrm{FB}}) \geq R(k^{\mathrm{FB}})$, the infimum of viable intensities is at most k^{FB}, while the choice of the platform is at least k^{FB}: drivers prefer weakly less speed than is efficient, and the platform weakly more;

(iii) under risk aversion without limited liability (Section 7), drivers get their outside option at every intensity and are indifferent to speed.

Proof. By Lemma 2 at intensity k, both types get the rent $R(k)$: type H gets $W_h^k - c(k) = W_n^k = R(k)$ with a binding care constraint, and type L gets $w_0 = R(k)$. Monotonicity is Theorem 3(i), and this gives (i). For (ii), the efficient intensity is viable by assumption, so the infimum of the viable intensities is at most k^{FB}, while $k^{\mathrm{SB}} \geq k^{\mathrm{FB}}$ by Theorem 3(ii). Part (iii) is Lemma 4 applied at each intensity. □

So speed pressure is a way to extract rent, and drivers oppose it even when speed is productive and even when they are careful. This fits with the protests of delivery workers against ratings and automatic penalties that are linked to speed (TechCrunch, 2026). It also gives a testable prediction: opposition to speed targets should be strongest where limited liability binds, that is among workers with little wealth, and weakest where pay can be adjusted freely.

5.4 Monitoring and accidents

Proposition 5 (Monitoring and aggregate accidents)**.** *Under the assumptions of Proposition 3:*

(i) Whenever Express operates, expected accidents per driver are $\phi\, k^{\mathrm{SB}} q_h$, which is weakly larger than the first-best level $\phi\, k^{\mathrm{FB}} q_h$ (strictly under the conditions of Theorem 3(iii)).

(ii) Suppose that without telematics care is implementable at no intensity, that is, $c(k)/k > \kappa(q_n - q_h)$ for all k, while with telematics Express operates at a second-best intensity k_T^{SB}. Then introducing telematics raises expected accidents from 0 to $\phi\, k_T^{\mathrm{SB}} q_h > 0$, even though in every critical episode a careful driver who takes care has a lower accident probability (q_h) than a careless one (q_n).

The proof follows directly from Theorem 3, Proposition 3, and Lemmas 2 and 3. Monitoring reduces accidents at the intensive margin (care in each episode), but it can increase them at the extensive margin. It creates Express work that would not exist otherwise, and this work runs faster than is efficient. So aggregate accident counts are not a good measure of whether monitoring "works". The effect is similar to Peltzman (1975), but it comes from contracting and not from risk compensation by drivers.

When Express operates both with and without telematics, telematics does not necessarily make the intensity distortion worse. Since it lowers rents, it can make $R(k)$ flatter and *reduce* the gap $k^{\mathrm{SB}} - k^{\mathrm{FB}}$. The link between monitoring and over-speeding is through the extensive margin: monitoring creates the markets where over-speeding happens.

5.5 What the device records matters

The dilution effect relies on the fact that the platform cannot distinguish an uneventful critical episode from a calm trip. But some modern systems (camera-based driver monitoring, near-miss detection) do record exposure.

Proposition 6 (Exposure-aware monitoring)**.** *Maintain Assumptions 1 and 2 and Assumption 3(ii), and suppose the device also records whether a critical episode happened. Then $R(k) = k\, R^{\mathrm{crit}}\big(c(k)/k\big)$, where*

$$R^{\mathrm{crit}}(\gamma) := \min\big\{W_n(x) : x \in \mathcal{X}(\kappa),\ W_h(x) - W_n(x) \geq \gamma\big\}.$$

(i) If $c(k) = k\gamma$ and $0 < R^{\mathrm{crit}}(\gamma) < \infty$, then R is strictly increasing, the implementable set does not depend on k, and every second-best intensity is weakly below every first-best intensity.

(ii) If $c(k) = c$ is constant, then R is nonincreasing in k. So when the first-best and second-best intensities are unique, $k^{\mathrm{SB}} \geq k^{\mathrm{FB}}$, and the distortion is not reversed.

Proof. When exposure is recorded, the schedule can pay x_{calm} on calm trips and $x(\omega)$ in critical episodes. The care constraint is $k[W_h(x) - W_n(x)] \geq c(k)$, and the rent is $(1-k)x_{\mathrm{calm}} + kW_n(x)$. It is optimal to set $x_{\mathrm{calm}} = 0$, and this gives $R(k) = kR^{\mathrm{crit}}(c(k)/k)$. (i) Here $R(k) = kR^{\mathrm{crit}}(\gamma)$, and the comparison of k^{SB} with k^{FB} is the argument of Theorem 3(ii) with the inequalities reversed. (ii) R^{crit} is convex, since it is the value of a linear program as a function of its right-hand side, and $R^{\mathrm{crit}}(0) = 0$. So $R^{\mathrm{crit}}(\gamma)/\gamma$ is nondecreasing, and $R(k) = c \cdot R^{\mathrm{crit}}(c/k)/(c/k)$ is nonincreasing in k. If $k^{\mathrm{SB}} < k^{\mathrm{FB}}$ and both are unique, then k^{FB} is implementable (the implementable set is an up-set), and the two strict optimality inequalities add up to $R(k^{\mathrm{SB}}) < R(k^{\mathrm{FB}})$, a contradiction. □

Together, Theorem 3 and Proposition 6 say that *what* a monitoring technology records, and not only how precise it is, shapes how much speed the platform wants. Event-based telematics, which records only adverse events, rewards the absence of bad news, and so it benefits from exposure. With proportional care costs, exposure-aware telematics rewards good handling of observed critical episodes, and it pushes the platform toward less speed.

6 Policy: speed caps, data rules and reporting

Rents are transfers, so total surplus (platform plus drivers) is $V_0 + \phi S(k)$ when Express operates at intensity k, and V_0 otherwise. Suppose also that each accident causes a harm $E \geq 0$ to third parties, which the platform does not internalize (Edlin and Karaca-Mandic, 2006). Then social surplus is $V_0 + \phi[S(k) - kq_hE]$ when Express operates. Let k^{soc} maximize $S(k) - kq_hE$. Since kq_hE is nondecreasing in k, the argument of Theorem 3(ii) gives $k^{\mathrm{soc}} \leq k^{\mathrm{FB}}$ whenever both maximizers are unique.

6.1 Speed caps

Suppose a regulator imposes a cap $k \leq \bar{k}$. This is basically what the Indian labor ministry did when it asked platforms to stop the ten-minute promises.

Proposition 7 (Speed caps)**.** *Maintain the assumptions of Proposition 3, and suppose S is strictly quasi-concave with peak k^{FB}. Consider a cap $\bar{k} \leq k^{\mathrm{FB}}$.*

(i) If $\bar{k} \in \mathcal{K}(\kappa)$ and $\phi S(\bar{k}) \geq R(\bar{k})$, the platform operates Express at exactly $k = \bar{k}$. Otherwise it shuts Express down.

(ii) A cap at $\bar{k} = k^{\mathrm{FB}}$ restores the first-best intensity, and weakly raises total surplus, whenever the platform stays active. If $S(k^{\mathrm{soc}}) - k^{\mathrm{soc}}q_hE \geq 0$, a cap at $\bar{k} = k^{\mathrm{soc}}$ implements the social optimum, under the same condition.

(iii) If the market stays open under a binding cap, the rent of each driver goes up from $R(k^{\mathrm{SB}})$ to $R(\bar{k})$. A cap that is too tight, with $\phi S(\bar{k}) < R(\bar{k})$, shuts down the market, and this lowers total surplus to V_0 and the rents of drivers to zero.

Proof. For $k \leq \bar{k} \leq k^{\mathrm{FB}}$ with $k \in \mathcal{K}(\kappa)$, strict quasi-concavity gives $S(k) \leq S(\bar{k})$, and $\bar{k} \in \mathcal{K}(\kappa)$ by the up-set property. Then Theorem 3(i) gives $R(k) \geq R(\bar{k})$. So $\phi S(k) - R(k) \leq \phi S(\bar{k}) - R(\bar{k})$, with strict inequality for $k < \bar{k}$, and this proves (i). Parts (ii) and (iii) follow from (i), the definitions of total and social surplus, and the monotonicity of R. □

This result goes against a natural idea. One may think that once a platform monitors driving behavior, speed regulation is not needed anymore, because monitoring already makes drivers careful. In my model, monitoring is what makes the Express market viable when hiding is cheap, and in this market the platform wants to push speed above the efficient level. So monitoring and speed caps are complements on the extensive margin: caps matter only where Express operates, and monitoring makes this set larger. (As I noted in Section 5, where Express operates anyway, better monitoring does not necessarily increase the intensity distortion.) Caps also redistribute to drivers, and this is consistent with the demands of drivers for relief from speed pressure. But caps must be moderate. The profit of the platform falls when the cap becomes tighter, and a cap that is too tight destroys the market that it wants to regulate.

6.2 Restricting the use of telematics

Privacy rules sometimes limit how monitoring data can be used in pay or discipline.

Proposition 8 (Data restrictions)**.** *In the binary-mode model, suppose a rule forbids conditioning pay on the telematics signal.*

(i) If $\kappa < \kappa_0$ and the market operated with telematics, the rule eliminates Express. Expected accidents fall from ϕq_h to 0, total surplus falls by ϕS, and social surplus changes by $-\phi(S - q_h E)$. So the rule raises social surplus if and only if $S < q_h E$.

(ii) If $\kappa \geq \kappa_0$ and the market operates under both regimes ($\phi S \geq R_0$), the rule raises the rents of drivers from $R_T(\kappa)$ to R_0. If instead $R_T(\kappa) \leq \phi S < R_0$, the rule closes the market and rents fall to zero.

Proof. This follows directly from Theorem 1, Corollary 2 and Lemma 2. □

So a data restriction works like a blunt speed limit. It is socially valuable only when the Express market should not exist at all, that is, when the harm to third parties is larger than the private surplus. In the critical-episode model, suppose hiding is so cheap that Express is feasible only with telematics. Then a ban gives social surplus V_0. A cap at k^{soc} that keeps the platform active gives $V_0 + \phi[S(k^{\mathrm{soc}}) - k^{\mathrm{soc}} q_h E]$, which is at least V_0 whenever Express is socially worthwhile. So when the market is worth having, it is better to target the intensity directly than to ban the information that makes the market possible.

6.3 Reporting protections and the cost of hiding

Policies like mandatory incident reporting, dashcam evidence and fines for unreported accidents raise κ. By Theorems 1 and 2, $R_T(\kappa)$ is nonincreasing and convex in κ, and raising κ above κ_0 creates the market even without telematics. So the cost of hiding and behavioral monitoring are *substitutes*. In the other direction, Lemma 1 implies that any contract that penalizes a reported accident by more than κ is dominated. Harsh penalties for reported accidents (for example automatic deactivation) do not work whenever hiding is cheaper than the penalty. They only turn accidents into unreported accidents, so the principal learns nothing and still pays the cost. The optimal penalty for a reported accident is never larger than κ. When this cap binds, the optimal contract pays a positive amount after a reported accident together with a clean telematics record (Proposition 1).

7 Risk aversion without limited liability

In this section I replace limited liability with risk aversion, which is the other standard source of agency costs, and I ask which results survive. The payoff of the driver is $u(x) - c\,a$, where u is strictly increasing, strictly concave and three times differentiable on an open interval $(\underline{x}, \infty)$, with $u(x) \to -\infty$ as $x \downarrow \underline{x}$. Examples are CARA, and CRRA with relative risk aversion $\sigma \geq 1$ and $\underline{x} = 0$ (log utility when $\sigma = 1$). There are no other restrictions on payments. Let $\psi := u^{-1}$, which is increasing and strictly convex, and let $\bar{u} = u(\bar{w})$ be the outside option. Hiding costs κ in money, so the truthful-reporting constraints are still $0 \leq x(0,s) - x(1,s) \leq \kappa$. The proof of Lemma 1 still works, as long as payments net of hiding costs stay in the domain of u.

7.1 Screening is redundant

Lemma 4 (Redundancy of the screening constraint)**.** *Under Assumption 1, consider the* relaxed *problem in which the principal induces care from H in Express, pays L the reservation wage $\bar{w}$ in Standard, and ignores the incentive constraint of L:*

$$C^{\mathrm{RA}}(\kappa) := \min_{x} \ \mathbb{E}_h[x] \quad s.t. \quad \mathbb{E}_h[u(x)] - \mathbb{E}_n[u(x)] \geq c, \quad \mathbb{E}_h[u(x)] - c \geq \bar{u}, \quad 0 \leq x(0,s) - x(1,s) \leq \kappa.$$

At any solution, (i) the participation constraint of H binds, and (ii) $\mathbb{E}_n[u(x)] \leq \bar{u}$, so L weakly prefers Standard. So the solution of the relaxed problem also solves the full problem. L gets no rent, and Express is optimal if and only if $V_1 - V_0 - q_h D \geq C^{\mathrm{RA}}(\kappa) - \bar{w}$, a condition that does not depend on ϕ.

Proof. (i) Suppose participation is slack. Then lower every utility level by $\varepsilon > 0$, that is, replace $x(\omega)$ by $\psi(u(x(\omega)) - \varepsilon)$. Utility differences do not change, so the care constraint holds. Monotonicity is also preserved. For the hiding constraint, $\psi(v_0 - \varepsilon) - \psi(v_1 - \varepsilon) \leq \psi(v_0) - \psi(v_1)$ whenever $v_0 \geq v_1$, because ψ is convex. Expected payments strictly fall, which is a contradiction. (ii) By the care

constraint and (i), $\mathbb{E}_n[u(x)] \le \mathbb{E}_h[u(x)] - c = \bar{u}$. The last claim compares $\phi(V_1 - q_h D - C^{\mathrm{RA}}) + (1-\phi)(V_0 - \bar{w})$ with $V_0 - \bar{w}$. Other regimes are dominated exactly as in Lemma 2. □

So under risk aversion without limited liability, adverse selection does not matter: the moral-hazard contract that holds H to her reservation utility automatically keeps L out. The reason is the one in Remark 2. Without limited liability, the principal can lower utility after every outcome until H is just willing to participate, and this pushes the mimicking payoff of the careless type down to her outside option. With limited liability the principal cannot pay less than zero, so mimicking is still profitable for the careless type. So the adverse-selection part of the lemons tax in Remark 1 comes from limited liability.

7.2 Which results survive

Proposition 9 (Speed limit under risk aversion)**.** *Without telematics, care in Express is implementable if and only if*

$$\kappa \;\ge\; \kappa_0^{\mathrm{RA}} := \psi\big(\bar{u} + c + q_h\Delta\big) - \psi\big(\bar{u} + c - (1-q_h)\Delta\big), \qquad \Delta := \frac{c}{q_n - q_h},$$

with $\kappa_0^{\mathrm{RA}} := \infty$ *if* $\bar{u} + c + q_h\Delta \ge \sup u$*. The threshold* κ_0^{RA} *is increasing in* $\bar{u}$*.*

Proof. In utils, let $v_0 := u(x(0))$ and $v_1 := u(x(1))$, and let $\delta := v_0 - v_1$. Care requires $\delta \ge \Delta$, and participation requires $v_0 \ge \bar{u} + c + q_h\delta$. The monetary spread $\psi(v_0) - \psi(v_0 - \delta)$ is increasing in v_0 (by convexity of ψ) and in δ (since ψ is increasing). So its minimum over the set defined by these two constraints is at $\delta = \Delta$ and $v_0 = \bar{u} + c + q_h\Delta$, and it is equal to κ_0^{RA}. Care is implementable iff this minimum is at most κ. If $\bar{u} + c + q_h\Delta \ge \sup u$, no contract satisfies care and participation. Monotonicity in $\bar{u}$ also follows from the convexity of ψ. □

So the speed limit survives, and there is a wealth effect: a driver who is paid more is harder to punish in utility terms, so her threshold is higher.

Proposition 10 (Market creation under risk aversion)**.** *Suppose telematics is informative* ($\lambda^* > 1$)*, and utility can go up enough above the outside option:*

$$\bar{u} + c + \frac{c\,(1 - P_h(s^*))}{P_h(s^*) - P_n(s^*)} < \sup u.$$

This condition holds automatically if u *is unbounded above, for example for log utility. Then care in Express is implementable for every* $\kappa \ge 0$*.*

Proof. Use a schedule that ignores accidents, $x(0,s) = x(1,s) = \psi(v_s)$, which satisfies the reporting constraints for every $\kappa \ge 0$. Let $v_s = \bar{u} + c + \beta\,(\mathbf{1}\{s = s^*\} - P_h(s^*))$ with $\beta := c/[P_h(s^*) - P_n(s^*)]$. Then $\mathbb{E}_h[v] = \bar{u} + c$ and $\mathbb{E}_h[v] - \mathbb{E}_n[v] = \beta[P_h(s^*) - P_n(s^*)] = c$. The lower utility level can be reached because u is unbounded below, and the upper level can be reached by the condition above. □

This condition is not trivial. With CARA utility, which is bounded above, a weak signal needs a utility premium after the best signal that may be more than what money can buy. Then the market can still be missing for small κ, even with telematics.

Intensity. Consider the critical-episode model of Section 5 without telematics, where the cost of care is measured in utils. Let $C(k)$ be the second-best cost of employing H at intensity k, let $C^{\mathrm{FB}}(k) := \psi(\bar{u} + c(k))$ be the first-best cost, and let $A(k) := C(k) - C^{\mathrm{FB}}(k)$ be the *agency cost*. Let $F(k) := V(k) - kq_hD - C^{\mathrm{FB}}(k)$ be the first-best objective.

Proposition 11 (Intensity under risk aversion)**.**

(a) Proportional cost of care, $c(k) = k\gamma$. *Suppose ψ' is convex (equivalently $3u''(x)^2 \geq u'(x)u'''(x)$, which holds for CARA and for CRRA) and $0 < q_h < 1/2$. Then (i) the implementable intensities form a (possibly empty) down-set of $(0, 1]$, that is, an implicit speed limit on intensity; (ii) $A(k)$ is strictly increasing; and (iii) every second-best intensity is weakly below every first-best intensity, and strictly below when k^{FB} is interior and F is differentiable at k^{FB}.*

(b) Fixed cost of care, $c(k) = c$. *Suppose $q_h > 0$. Then the implementable intensities form an up-set, $A(k)$ is strictly decreasing, and every second-best intensity is weakly above every first-best intensity, and strictly above when k^{FB} is interior and F is differentiable at k^{FB}.*

Proof sketch. The full proof is in Appendix A. (a) As in Proposition 9, the optimal contract has utility spread $\Delta_\gamma := \gamma/(q_n - q_h)$ and top utility $v_0(k) = \bar{u} + k(\gamma + q_h\Delta_\gamma)$, which is increasing in k. So the monetary spread increases in k, and this proves (i). The agency cost is the Jensen gap of a two-point lottery with spread Δ_γ, low-outcome probability $kq_h < 1/2$, and mean $\bar{u} + k\gamma$. Convexity of ψ' implies that this gap increases in both the probability and the mean, which proves (ii). Part (iii) follows as in Theorem 3(ii), with the inequalities reversed. (b) The required utility spread is $c/[k(q_n - q_h)]$, which falls with k, while the top utility does not depend on k. So both the monetary spread and the Jensen gap fall with k. □

With proportional costs, which isolate the dilution effect of Section 5, the two frictions push in opposite directions. Under risk aversion, exposure to critical episodes is exposure to risk that must be compensated, so the platform uses less of it. Under limited liability, exposure is a source of information that cuts rents, so the platform wants more of it. With a fixed cost of care, the leverage effect pushes toward speed under both frictions. So the over-speeding from dilution in Proposition 3 is specific to limited liability, but the leverage effect is not. Figure 3 compares the first-order conditions in the proportional case. I also did numerical experiments with binary telematics, CARA utility and $c(k) = k\gamma$, and the agency cost was increasing in k in every parameterization I tried. So the reversal with proportional costs does not seem to depend on the absence of telematics, but I leave a general proof for future work. Table 1 summarizes the comparison.

Remark 4 (Screening without limited liability). A common way to write this problem is to use a risk-averse driver, no limited liability and the cognitive wall, and then ask whether the penalty

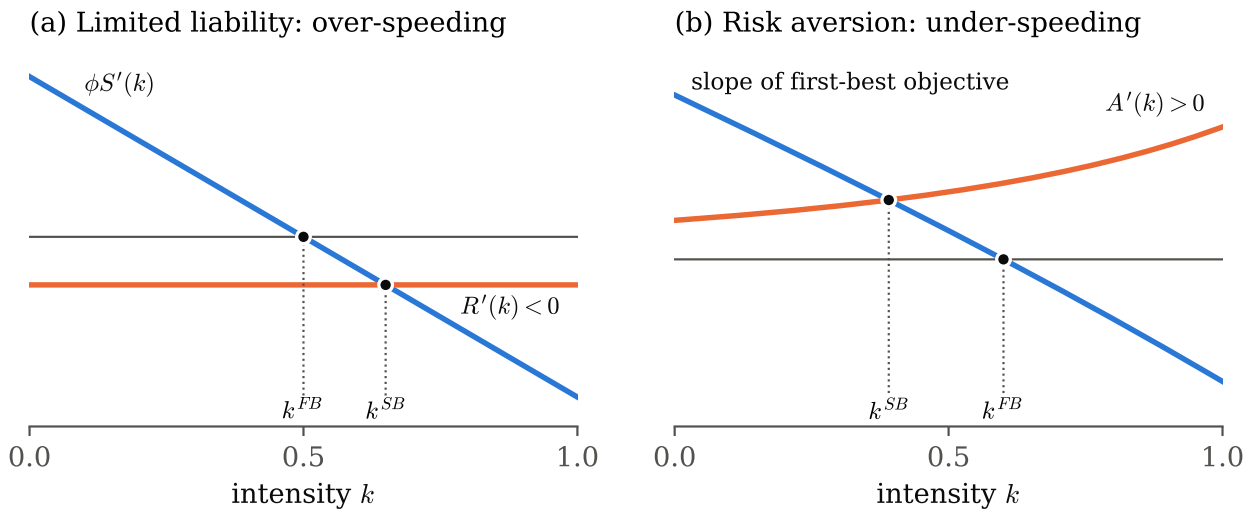


Figure 3: First-order conditions for intensity with a proportional cost of care, $c(k) = k\gamma$, and quadratic $V(k)$. Panel (a): limited liability, binary telematics ($q_h = 0.05$, $q_n = 0.15$, $\pi_h = 0.1$, $\pi_n = 0.7$, $\gamma = 1$, $\phi = 0.8$, κ large enough not to bind, $k^{\text{FB}} = 0.5$). The rent falls linearly in k, so k^{SB} solves $\phi S'(k) = R'(k) < 0$ and is to the right of k^{FB}. Panel (b): CARA utility with coefficient 1, $\bar{w} = 1$, $\gamma = 0.3\,|\bar{u}|$, no telematics ($q_h = 0.1$, $q_n = 0.3$, $D = 2$, κ large enough not to bind, $k^{\text{FB}} = 0.6$). The agency cost goes up in k, so k^{SB} solves $F'(k) = A'(k) > 0$ and is to the left of k^{FB}. The horizontal line is zero.

needed to keep L out is larger than the penalty needed to motivate H. Lemma 4 shows that this cannot happen at an optimal contract. Once H is held to her reservation utility, keeping L out is implied by motivating H. The speed limit then comes only from moral hazard and hiding, through κ_0^{RA}. It is limited liability that gives adverse selection its own role.

8 Stress tests with hidden failures

In the static model, the information from Express is used only to pay for Express. In practice, platforms and insurers also use a first monitored period to decide what comes later, for example access to premium orders, a long-term contract, or a discount for the whole life of a policy. In this section I study a two-period version where Express is a probation stress test, and I ask how hidden failures change what such a test can reveal. I focus on the information motive, so there is no moral hazard during probation.

8.1 Setup

The type of a new driver $\theta \in \{H, L\}$ is persistent, with prior ϕ, and it is *unknown to both parties*: a newcomer does not yet know whether she can handle speed. In period 1 the principal either keeps the driver in Standard or puts her on *probation* in Express. Relative to Standard, probation gives the principal σ_θ (productivity net of expected accident costs), with $\sigma_L < 0$. During probation, behavior is determined by type: outcomes (y, s) are drawn from $f_H := f_h$ or $f_L := f_n$. As before,

Table 1: Limited liability versus risk aversion.

	Limited liability, risk neutral (Sections 3 to 5)	Risk averse, no limited liability (Section 7)
Screening constraint	binds; careless type gets rent R (Lemma 2)	redundant; careless type gets no rent (Lemma 4)
Role of ϕ	market needs $\phi S \geq R$ (lemons tax)	irrelevant
Speed limit (no telematics)	$\kappa < \kappa_0 = c/(q_n - q_h)$	$\kappa < \kappa_0^{\mathrm{RA}}$, increasing in $\bar{u}$
Telematics creates market	yes (Corollary 2)	yes, if utility can go up enough (Proposition 10)
Accident penalty	in $[0, \kappa]$; leniency forced (Proposition 2)	in $[0, \kappa]$
Drivers and speed	payoff $R(k)$ falls with k: oppose speed (Proposition 4)	outside option at every k: indifferent
Distortion if $c(k) = k\gamma$	upward (Proposition 3)	downward (Proposition 11(a))
Distortion if $c(k) = c$	upward (Proposition 3)	upward (Proposition 11(b))

accidents can be hidden at cost κ and claimed at no cost.

In period 2 the principal gives a *license* (premium access) with probability $\tau(\hat{y}, s) \in [0, 1]$. A licensed H driver is worth $G_H > 0$ to the principal, and a licensed L driver costs $G_L > 0$. Having the license is worth $r > 0$ to the driver (this is a rent, not a transfer from the principal). The principal can also make transfers $x(\hat{y}, s) \geq 0$, and she commits to (τ, x) in advance. Define

$$g(\omega) := \phi f_H(\omega) G_H - (1 - \phi) f_L(\omega) G_L, \qquad \bar{f} := \phi f_H + (1 - \phi) f_L, \qquad \bar{\sigma} := \phi \sigma_H + (1 - \phi) \sigma_L.$$

Without probation, the principal licenses everyone or no one and gets $\Gamma_0 := \max\{0, \phi G_H - (1 - \phi) G_L\}$. I assume that f_L has full support and that accidents are bad news during probation: $f_H(1, s)/f_L(1, s) \leq f_H(0, s)/f_L(0, s)$ for every s.

Lemma 5 (Truthful reporting in probation)**.** *It is without loss to restrict attention to policies that satisfy, for every s,*

$$0 \;\leq\; \big[x(0, s) + r\tau(0, s)\big] - \big[x(1, s) + r\tau(1, s)\big] \;\leq\; \kappa.$$

Proof. Fix a policy and the optimal reports of the driver. If a driver would hide at $(1, s)$, set $\tau'(1, s) := \tau(0, s)$ and $x'(1, s) := \max\{0, x(0, s) - \kappa\}$. If she would fabricate at $(0, s)$, set $(\tau', x')(0, s) := (\tau, x)(1, s)$. Hiding and fabricating cannot both be optimal after the same signal. The new policy satisfies the constraints above, gives the same license decisions, weakly lowers every payment, and keeps every payoff nonnegative. Since the driver makes no choice before this, nothing else needs to be preserved. □

8.2 The optimal probation

Proposition 12 (Optimal probation)**.** *The value of probation is* $\Pi_P = \bar{\sigma} + \sum_s v_s$*, where, writing* $g_y := g(y,s)$*,*

$$v_s = \max\{0,\ g_0 + g_1\} + I_s(r,\kappa),$$

$$I_s(r,\kappa) = \begin{cases} \max\Big\{\alpha_s \min\big(1, \tfrac{\kappa}{r}\big),\ \alpha_s - \bar{f}(1,s)\,(r-\kappa)^+\Big\} & \text{if } g_0 > 0 > g_1, \\ 0 & \text{otherwise,} \end{cases}$$

and $\alpha_s := \min\{g_0, -g_1\}$ *measures the value of discriminating by accidents after signal* s*. When* $g_0 > 0 > g_1$*, the optimal policy has one of two forms. Either it* forgives*: the license probabilities after clean and accident records differ by at most* κ/r*, and there are no transfers. Or it* buys the truth*: it revokes the license after a reported accident and pays the reporter* $(r-\kappa)^+$*. It forgives if* $\alpha_s \le \bar{f}(1,s)\,r$ *and buys the truth otherwise.*

The proof is in Appendix A. It uses the fact that the program separates across signals, and then it writes the problem for each signal in terms of the license gap $\delta := \tau(0,s) - \tau(1,s)$. The accident information is worth $\alpha_s\delta$, and the reporting constraint requires compensation $(r\delta - \kappa)^+$.

Proposition 13 (Goodhart's law for self-reported tests)**.**

(i) Without telematics, $\Pi_P - \Gamma_0 = \bar{\sigma} + I_0(r,\kappa)$*. If* $g(0) > 0 > g(1)$*, then with* $\alpha := \min\{g(0), -g(1)\}$ *we have* $I_0(r,\kappa) = \alpha\kappa/r$ *for all* $r \ge \max\{\kappa, \alpha/\bar{f}(1)\}$*; otherwise* $I_0 \equiv 0$*. So the value of the test goes to zero as the stakes* r *grow relative to* κ*.*

(ii) With telematics, $\Pi_P - \Gamma_0 \ge \bar{\sigma} + I_T$*, where*

$$I_T := \sum_s \max\big\{0,\ g(0,s) + g(1,s)\big\} - \max\Big\{0,\ \sum_s \big[g(0,s) + g(1,s)\big]\Big\} \ \ge\ 0$$

does not depend on r *or* κ*.*

(iii) Testing for its own sake. *If* $\bar{\sigma} < 0 < \bar{\sigma} + I_T$*, the principal runs the Express probation with telematics for every* r*, even though probation is unprofitable by itself (possibly even for careful drivers,* $\sigma_H < 0$*). Without telematics she stops probation for all* $r > \max\{\kappa,\ \alpha/\bar{f}(1),\ \alpha\kappa/|\bar{\sigma}|\}$ *if* $g(0) > 0 > g(1)$*, and for all* r *otherwise.*

Proof. (i) With a singleton signal, $\sum_\omega g(\omega) = \phi G_H - (1-\phi)G_L$, so $\max\{0, g_0 + g_1\} = \Gamma_0$. If $r \ge \kappa$ and $\alpha \le \bar{f}(1)r$, then $\alpha\kappa/r \ge \alpha - \bar{f}(1)(r-\kappa)$, so $I_0 = \alpha\kappa/r$. (ii) Licensing on the signal alone, $\tau(0,s) = \tau(1,s) = \mathbf{1}\{g(0,s) + g(1,s) > 0\}$ with $x = 0$, satisfies Lemma 5 for every r and κ. (iii) follows from (i) and (ii). □

The intuition is simple. When the result of a test decides access to a valuable prize, the tested person cares about the record, and a self-reported record bends. The principal can make the prize depend less on reported accidents, but then the test is blunt. Or she can pay drivers to report accidents that cost them the prize, but this is expensive. In both cases, a high-stakes test that is

based on self-reports is worth little. A test based on behavioral data that cannot be manipulated does not have this problem. So a platform with telematics can find it worthwhile to put newcomers through a risky probation whose only purpose is to learn who can handle speed. Without telematics, a platform that wants to learn from accidents must keep the stakes of each single test low relative to κ, for example by giving privileges in small steps.

Remark 5. If drivers knew their types, the principal could offer a self-selection menu with a probation track and a Standard track. As in Lemma 2, L would then have to be paid what she expects from the probation track, $r\,\mathbb{E}_{f_L}[\tau] + \mathbb{E}_{f_L}[x]$. The license rule would then trade off G_H against this rent, and not against G_L. The effect of hiding on the informativeness of accident reports is the same. But truthful reporting would have to be shown separately, because the replacement used in Lemma 5 can raise the payoff of the driver, and this matters once she makes a self-selection choice.

9 Conclusion

Risky work produces failures, and workers can often hide them. In this paper I show that this simple fact changes the economics of risky work. When failures are the only evidence, a productive risky market can be missing completely (an implicit speed limit). When the market exists, careless workers must be paid to stay out of it (a lemons tax). Behavioral evidence that cannot be manipulated creates the market. The optimal contract then follows a pecking order by informativeness, and it tolerates reported failures only because otherwise they would be hidden. The platform would pay to make failures harder to hide. When the principal also chooses how much risk the work has, exposure becomes a screening device. A critical episode that is handled well looks like a calm one, so rents fall with exposure. Therefore a principal who faces workers with limited liability puts more risk on them than is efficient, and the workers oppose it. This force depends on what the monitoring device records, and when the cost of care scales with exposure, it reverses under pure risk aversion. Finally, tests based on self-reported failures lose their value when the stakes go up, but tests based on behavioral evidence do not.

These results complement the work on motivating agents to take risk (Manso, 2011; Chade and Swinkels, 2026). In that work the agent chooses risk, and the principal may treat failure leniently to encourage it. In my model the principal chooses risk to learn about the agent, and leniency toward failure is a constraint that comes from the ability of the agent to hide failures. For policy, the model says three things. Monitoring and speed regulation are complements, because monitoring creates the markets where over-speeding happens. Making failures harder to hide is a substitute for behavioral surveillance. And punishing reported accidents by more than the hiding cost does not work.

Toward a dynamic model. The static results suggest a continuous-time version, similar to Biais et al. (2010) and Sannikov (2008). Accidents arrive as Poisson events with intensity $k\,q_j$, telematics is a diffusion whose drift depends on care and whose informativeness scales with exposure k, and

the contract is written on the continuation value U_t of the driver. The hiding constraint becomes a bound on the *jump* in continuation value at a reported accident, $U_t - U_{t^-} \geq -\kappa$, and the care constraint requires the sensitivity of U_t to accidents and telematics to be larger than the flow cost of care. I think two questions are especially interesting. First, when the jump is bounded by κ, does the optimal contract move incentives from accident-based jumps to telematics-based diffusion exposure as U_t gets close to the limited-liability boundary? Second, does the optimal speed path of the platform front-load intensity, that is, test new drivers hard and then relax, as the probation analysis suggests?

Empirical implications. The model predicts that (i) after behavioral monitoring is introduced, high-speed services should enter or expand, and not only driving within existing services should become safer; (ii) accident penalties in platform contracts should be bounded by proxies for hiding costs (for example, they should be lower for small incidents that are easy to hide); (iii) where the cost of care scales with exposure, the pressure to over-speed, and the opposition of workers to it, should be strongest where workers are most liquidity-constrained; and (iv) monitoring systems that record exposure (near misses, context) should come with less speed pressure than event-only systems, especially where the cost of care scales with exposure.

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A Proofs

Proof of Lemma 1

Fix s. Under x, a driver at $(0, s)$ can report truthfully (payoff $x(0, s)$) or claim an accident (payoff $x(1, s)$). A driver at $(1, s)$ can report truthfully (payoff $x(1, s)$) or hide the accident (payoff $x(0, s) - \kappa$). So her optimal payoffs are $\hat{x}(0, s)$ and $\hat{x}(1, s)$, and this proves (ii).

For (i), $\hat{x} \geq 0$ is immediate. Next, $\hat{x}(0, s) \geq \hat{x}(1, s)$, because $\max\{x(0, s), x(1, s)\}$ weakly exceeds both $x(1, s)$ and $x(0, s)-\kappa$. Finally, $\hat{x}(0, s)-\hat{x}(1, s) \leq \kappa$. If $x(0, s) \geq x(1, s)$, then $\hat{x}(0, s) = x(0, s) \leq \hat{x}(1, s) + \kappa$ since $\hat{x}(1, s) \geq x(0, s) - \kappa$. If $x(1, s) > x(0, s)$, then $\hat{x}(0, s) = x(1, s) \leq \hat{x}(1, s)$.

For (iii), under x the principal pays the amount attached to the report. At $(0, s)$ this is $\max\{x(0, s), x(1, s)\} = \hat{x}(0, s)$, because a fabricating driver receives $x(1, s)$ and incurs no cost. At $(1, s)$ it is $x(1, s) = \hat{x}(1, s)$ if the driver reports, and $x(0, s) = \hat{x}(1, s) + \kappa$ if she hides it. The payoff of the driver at every outcome does not change, so the expected payoff of every profile does not change, and the care and selection decisions also do not change. Reporting does not affect true accidents, so the gross payoff of the principal also does not change. □

Proof of Lemma 2

By Lemma 1, consider a direct mechanism in which each reported type is assigned a mode and a payment rule, with Express schedules in $\mathcal{X}(\kappa)$. Let $U_\theta \geq 0$ denote type θ's equilibrium payoff. The principal's expected payment to θ equals U_θ, plus c if $\theta = H$ takes care.

Step 1: mechanisms without care earn at most V_0. If H is not assigned Express with care, every driver who works generates gross value V_0 (Standard) or $V_1 - q_n D < V_0$ (Express without care, by Assumptions 1 and 2), and a driver who does not work generates $0 \leq V_0$. Payments are nonnegative, so profit is at most V_0.

Step 2: an upper bound with care. Suppose H is assigned Express with schedule x_H and takes care. Optimality of care gives $W_h(x_H) - c \geq W_n(x_H)$, so x_H is feasible in (1) and $W_n(x_H) \geq R(\kappa)$. Type L can report H and obtain $W_n(x_H)$, so $U_L \geq W_n(x_H)$. If L is assigned Standard, the principal's profit from L is $V_0 - U_L \leq V_0 - W_n(x_H)$. If L is assigned Express, it is $V_1 - q_n D - U_L < V_0 - W_n(x_H)$. The profit from H is $V_1 - q_h D - W_h(x_H) \leq V_1 - q_h D - c - W_n(x_H)$. Hence expected profit is at most

$$\phi(V_1 - q_h D - c) + (1 - \phi)V_0 - W_n(x_H) = V_0 + \phi S - W_n(x_H) \leq V_0 + \phi S - R(\kappa).$$

Step 3: attainment. If $R(\kappa) < \infty$, (1) is a linear program with a nonempty feasible set and an objective bounded below by zero, so it has a solution x^*. If $W_h(x^*) - W_n(x^*) > c$, then tx^* with $t := c/[W_h(x^*) - W_n(x^*)] \in (0,1)$ is feasible, since $\mathcal{X}(\kappa)$ is closed under scaling by $t \in [0,1]$, and has weakly lower rent. So there is a solution at which the care constraint binds. Under the menu $(W_n(x^*), x^*)$, type H obtains $W_n(x^*)$ from Express with care, from Express without care, and from Standard, and she follows the recommendation. Type L obtains $W_n(x^*)$ from either mode and follows the recommendation of Standard. Profit equals $V_0 + \phi S - R(\kappa)$. Assigning both types to Standard at wage zero attains V_0.

Lotteries and ties. If a mechanism assigns H to Express with care with probability α and otherwise to an allocation covered by Step 1, the same arguments bound profit by $V_0 + \alpha(\phi S - R(\kappa))$, which is maximized at $\alpha \in \{0,1\}$. At the optimal menu both types are indifferent between options, and a driver with $d_s = \kappa$ is indifferent between reporting and hiding. Tie-breaking in the principal's favor affects only whether the optimum is attained, except at knife-edges where care is feasible only with every premium at the cap (e.g., $\kappa = \kappa_0$ without telematics). Otherwise, lowering capped premia to $\kappa - \varepsilon$, raising an uncapped instrument so that $W_h - W_n = c + \varepsilon$, and choosing w_0 strictly between W_n and $W_h - c$ yields strict incentives at a profit within $O(\varepsilon)$ of the bound. □

Proof of Theorem 2

Step 1: reformulation. The map $x \mapsto (b,d)$ with $b_s = x(1,s)$ and $d_s = x(0,s) - x(1,s)$ is a bijection from $\mathcal{X}(\kappa)$ onto $\{(b,d) : b \geq 0,\ 0 \leq d \leq \kappa\}$. For each profile, $W_j(x) = \sum_s [P_j(s) b_s + f_j(0,s) d_s]$. Substituting gives the knapsack with the yields and rent costs in the table. For each instrument, ρ_i is the probability under n of the event on which it pays and $\rho_i + m_i$ is the corresponding probability under h. Hence $\mathrm{LR}_i = (\rho_i + m_i)/\rho_i$ and $m_i/\rho_i = \mathrm{LR}_i - 1$.

Step 2: the greedy solution is optimal. Instruments with $m_i \leq 0$ can be set to zero without loss. Order the remaining ones by decreasing likelihood ratio, and let z^* fill them in this order until $\sum_i m_i z_i = c$. If the order is exhausted first, the maximal yield $\sum_{m_i > 0} m_i \,\mathrm{cap}_i$ is below c and the program is infeasible. Otherwise, let $i^\dagger$ be the marginal instrument and set

$$\zeta := \rho_{i^\dagger}/m_{i^\dagger} = 1/(\mathrm{LR}_{i^\dagger} - 1) \geq 0, \qquad \eta_i := (\zeta m_i - \rho_i)^+ \text{ for capacity-limited } i.$$

The dual program is max $\zeta c - \kappa \sum_{i \text{ limited}} \eta_i$ subject to $\zeta m_i - \eta_i \leq \rho_i$ for limited i, $\zeta m_i \leq \rho_i$ for unlimited i, and $\zeta, \eta \geq 0$.

Dual feasibility. For limited instruments it holds by construction. For an unlimited instrument with $m_i > 0$, if $\mathrm{LR}_i > \mathrm{LR}_{i^\dagger}$, it would be filled before $i^\dagger$ and, having unlimited capacity, would complete the care constraint, contradicting the definition of $i^\dagger$. Hence $\mathrm{LR}_i \leq \mathrm{LR}_{i^\dagger}$, which is equivalent to $\zeta m_i \leq \rho_i$.

Complementary slackness. Instruments filled before $i^\dagger$ have $\mathrm{LR}_i \geq \mathrm{LR}_{i^\dagger}$, hence $\eta_i = \zeta m_i - \rho_i$ and a binding dual constraint, and they are filled to capacity. The marginal instrument satisfies $\zeta m_{i^\dagger} = \rho_{i^\dagger}$ with $\eta_{i^\dagger} = 0$. Instruments after $i^\dagger$ have $z_i^* = 0$ and $\eta_i = 0$. The care constraint binds.

Hence z^* is optimal. If $\mathrm{LR}_{i\dagger} = \infty$, then $\zeta = 0$ and both values are zero.

Step 3: rent formula and properties. $R(\kappa) = \sum_i \rho_i z_i^* = \sum_i m_i z_i^*/(\mathrm{LR}_i - 1)$. As the cumulative yield t runs from 0 to c, the integrand equals $1/(\mathrm{LR}_i - 1)$ on the segment of length $m_i z_i^*$ that belongs to instrument i. Raising κ enlarges the capacities of limited instruments without changing the order, so $\Lambda(t)$ weakly increases pointwise and R weakly falls. R is the value of a linear minimization program whose right-hand side is affine in κ, and such values are convex in the right-hand side.

Step 4: parts (ii) and (iii). If $\lambda^* > 1$, the base bonus b_{s^*} is an unlimited instrument with positive yield, so the greedy procedure always terminates and $R(\kappa) < \infty$. Instruments with likelihood ratio below λ^* come after b_{s^*} and are never used. Ties among base bonuses with $\lambda_s = \lambda^*$ can be resolved in favor of s^*. The only instruments preceding b_{s^*} are premia with $\mu_s > \lambda^*$. If $\lambda^* = 1$, every base bonus has zero yield. The only instruments with positive yield are the premia with $\mu_s > 1$, whose total capacity yield is $\kappa \sum_{\mu_s > 1} [f_h(0, s) - f_n(0, s)]$. With a singleton signal this equals $\kappa(q_n - q_h)$. □

Proof of Proposition 11

At intensity k without telematics, a contract specifies utilities (v_0, v_1) after a clean record and after a reported accident. Calm trips and accident-free critical episodes both yield a clean record. With cost of care $c(k)$, the care constraint is $k(q_n - q_h)(v_0 - v_1) \geq c(k)$ and participation is $v_0 - kq_h(v_0 - v_1) - c(k) \geq \bar{u}$. The cost is $(1 - kq_h)\psi(v_0) + kq_h\psi(v_1)$, and the reporting constraint is $0 \leq \psi(v_0) - \psi(v_1) \leq \kappa$. By the argument of Lemma 4, the careless type's incentive constraint is implied, and participation binds (downward shift). The driver's utility is then a two-point lottery with mean $\bar{v} := \bar{u} + c(k)$, spread $\delta := v_0 - v_1$, and low-outcome probability $p := kq_h$. For fixed $\bar{v}$ and p, raising δ is a mean-preserving spread, so the cost is increasing in δ. The care constraint therefore binds: $\delta(k) = c(k)/[k(q_n - q_h)]$ and $v_0(k) = \bar{u} + c(k) + kq_h\,\delta(k)$. Among contracts satisfying care and participation, the monetary spread $\psi(v_0) - \psi(v_0 - \delta)$ is increasing in δ and in v_0, and participation requires $v_0 \geq \bar{u} + c(k) + kq_h\delta$. So the minimal spread is attained by this contract and equals $\chi(k) := \psi(v_0(k)) - \psi(v_0(k) - \delta(k))$. Care is implementable iff $\chi(k) \leq \kappa$ and $v_0(k) < \sup u$, and the contract above is then optimal.

Part (a): $c(k) = k\gamma$. Here $\delta(k) = \Delta_\gamma$ is constant and $v_0(k) = \bar{u} + k(\gamma + q_h\Delta_\gamma)$ is increasing, so χ is nondecreasing and the implementable set is a down-set, proving (i). For (ii), write $A(k) = J(p(k), \bar{v}(k))$ with $p(k) = kq_h$, $\bar{v}(k) = \bar{u} + k\gamma$, and

$$J(p, \bar{v}) := (1 - p)\,\psi(v^+) + p\,\psi(v^-) - \psi(\bar{v}), \qquad v^+ := \bar{v} + p\Delta_\gamma, \quad v^- := \bar{v} - (1 - p)\Delta_\gamma.$$

Since $\bar{v} = (1-p)v^+ + pv^-$ and ψ' is convex, Jensen's inequality gives $J_{\bar{v}} = (1-p)\psi'(v^+) + p\psi'(v^-) - \psi'(\bar{v}) \geq 0$. Next,

$$J_p = \psi(v^-) - \psi(v^+) + \Delta_\gamma\big[(1 - p)\psi'(v^+) + p\psi'(v^-)\big].$$

By the Hermite–Hadamard inequality for the convex function ψ', $\psi(v^+) - \psi(v^-) = \int_{v^-}^{v^+} \psi'(t)\,dt \leq$

$\Delta_\gamma[\psi'(v^+)+\psi'(v^-)]/2$. Hence

$$J_p \;\geq\; \Delta_\gamma\Big(\tfrac{1}{2}-p\Big)\big[\psi'(v^+)-\psi'(v^-)\big] \;>\; 0,$$

because $p = kq_h < 1/2$ and ψ is strictly convex. Therefore $A'(k) = q_h\, J_p + \gamma\, J_{\bar{v}} > 0$ since $q_h > 0$. For (iii), the principal maximizes $F(k) - A(k)$ over the implementable set (up to constants), whereas the first best maximizes $F(k)$. Suppose $k^{\mathrm{SB}} > k^{\mathrm{FB}}$. Then k^{FB} is implementable, because the set is a down-set. Optimality gives $F(k^{\mathrm{SB}}) - A(k^{\mathrm{SB}}) \geq F(k^{\mathrm{FB}}) - A(k^{\mathrm{FB}})$ and $F(k^{\mathrm{FB}}) \geq F(k^{\mathrm{SB}})$, so $A(k^{\mathrm{SB}}) \leq A(k^{\mathrm{FB}})$, which contradicts (ii). For strictness, if k^{FB} is not implementable then $k^{\mathrm{SB}} < k^{\mathrm{FB}}$ trivially. Otherwise $F'(k^{\mathrm{FB}}) = 0 < A'(k^{\mathrm{FB}})$, so slightly lower intensities, which are implementable, do strictly better.

Part (b): $c(k) = c$. Now $\delta(k) = c/[k(q_n - q_h)]$ is decreasing and $v_0 = \bar{u} + c + q_h c/(q_n - q_h)$ does not depend on k. Hence χ is decreasing and the implementable set is an up-set. Writing $v_1(k) := v_0 - \delta(k)$, we have $v_1'(k) = \delta(k)/k$ and $A(k) = (1-kq_h)\psi(v_0) + kq_h\psi(v_1(k)) - \psi(\bar{u}+c)$, so

$$A'(k) = -q_h\big[\psi(v_0) - \psi(v_1(k)) - \psi'(v_1(k))\,(v_0 - v_1(k))\big] < 0$$

by strict convexity of ψ. The comparison of k^{SB} with k^{FB} is the argument of part (a)(iii) with the inequalities reversed, using the up-set property. □

Proof of Proposition 12

By Lemma 5, the principal maximizes $\sum_\omega [g(\omega)\tau(\omega) - \bar{f}(\omega)x(\omega)]$ subject to the reporting constraints. Both the objective and the constraints separate across signals, so fix s and write $g_y = g(y,s)$, $\bar{f}_y = \bar{f}(y,s)$. The assumption that accidents are bad news implies that $g_1 > 0 \Rightarrow g_0 > 0$.

Upper bound. Take any feasible $(\tau_0, \tau_1, x_0, x_1)$ and let $\delta := \tau_0 - \tau_1$. The upper reporting constraint and $x \geq 0$ give $x_1 \geq (r\delta - \kappa)^+$, so the objective is at most $g_0\tau_0 + g_1\tau_1 - \bar{f}_1(r\delta - \kappa)^+$. If $g_0, g_1 \geq 0$ the objective is at most $g_0 + g_1$. If $g_0, g_1 \leq 0$ it is at most 0. Now let $g_0 > 0 > g_1$.

- If $\delta \leq 0$, then $g_0\tau_0 + g_1\tau_1 = (g_0+g_1)\tau_0 + g_1(\tau_1 - \tau_0) \leq \max\{0, g_0 + g_1\}$.
- If $\delta > 0$, then $g_0\tau_0 + g_1\tau_1 = (g_0+g_1)\tau_0 - g_1\delta$ with $\tau_0 \in [\delta, 1]$, which is at most $\max\{0, g_0 + g_1\} + \alpha_s\delta$. (Take $\tau_0 = 1$ if $g_0 + g_1 \geq 0$, where $\alpha_s = -g_1$; and $\tau_0 = \delta$ otherwise, where $\alpha_s = g_0$.)

Hence the objective is at most $\max\{0, g_0+g_1\} + \max_{\delta\in[0,1]}\big[\alpha_s\delta - \bar{f}_1(r\delta-\kappa)^+\big]$. The maximand is concave and piecewise linear in δ, with slope α_s on $[0, \min(1, \kappa/r)]$ and slope $\alpha_s - \bar{f}_1 r$ beyond. Its maximum is $\alpha_s \min(1, \kappa/r)$ if $\alpha_s \leq \bar{f}_1 r$, and $\alpha_s - \bar{f}_1(r-\kappa)^+$ otherwise. In each case this equals $I_s(r,\kappa)$.

Attainment. If $g_0, g_1 \geq 0$, set $\tau \equiv 1$ and $x \equiv 0$. If $g_0, g_1 \leq 0$, set $\tau \equiv 0$ and $x \equiv 0$. If $g_0 > 0 > g_1$, let δ^* be the maximizer above: $\delta^* = \min(1, \kappa/r)$ if $\alpha_s \leq \bar{f}_1 r$ (forgiveness), and $\delta^* = 1$ otherwise (buying the truth). Set $\tau_0 = 1$, $\tau_1 = 1 - \delta^*$ if $g_0 + g_1 \geq 0$, and $\tau_0 = \delta^*$, $\tau_1 = 0$ otherwise. Set $x_0 = 0$ and $x_1 = (r\delta^* - \kappa)^+$. The reporting difference equals $r\delta^* - x_1 = \min\{r\delta^*, \kappa\} \in [0, \kappa]$, and the objective attains the bound. □